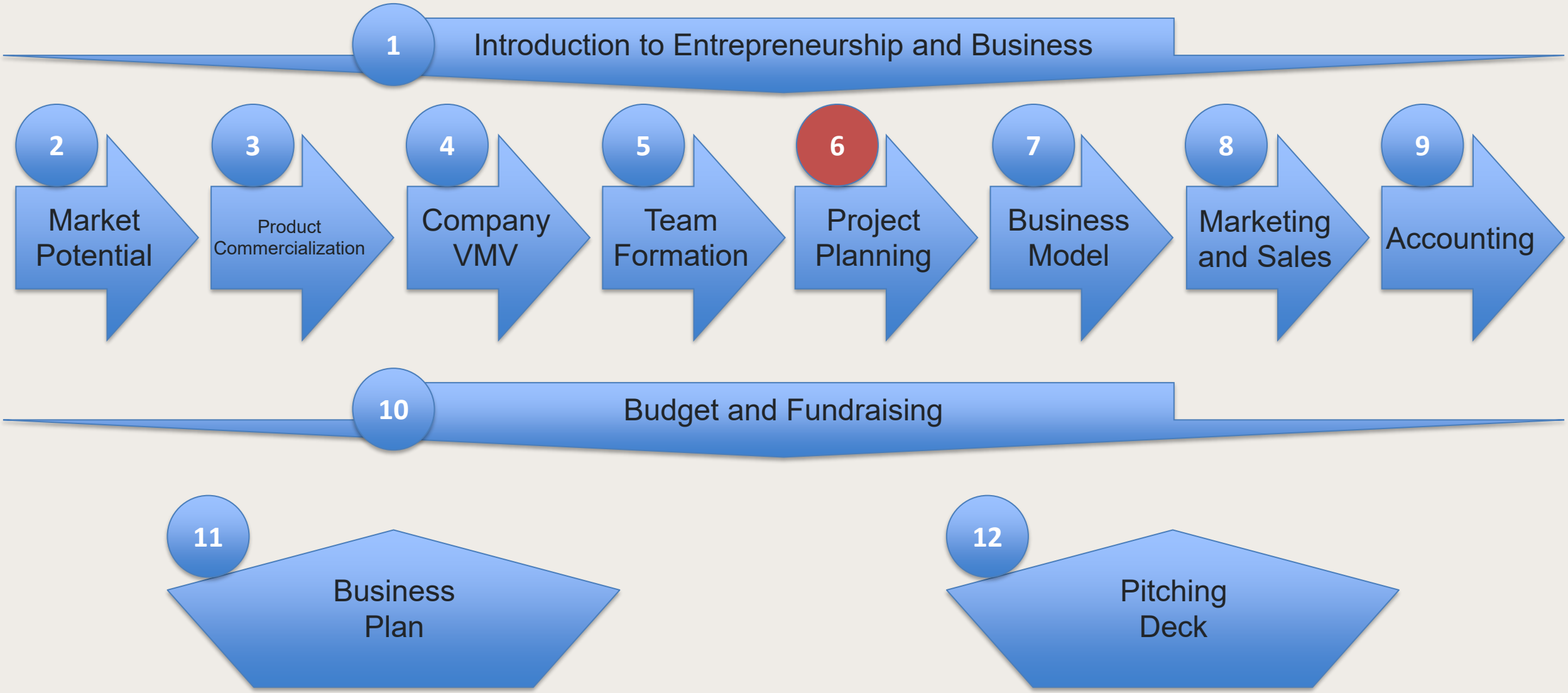
Strategic Product Innovation and Development for Startups

Ir Prof Alan Lam, JP



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Outline of the Course



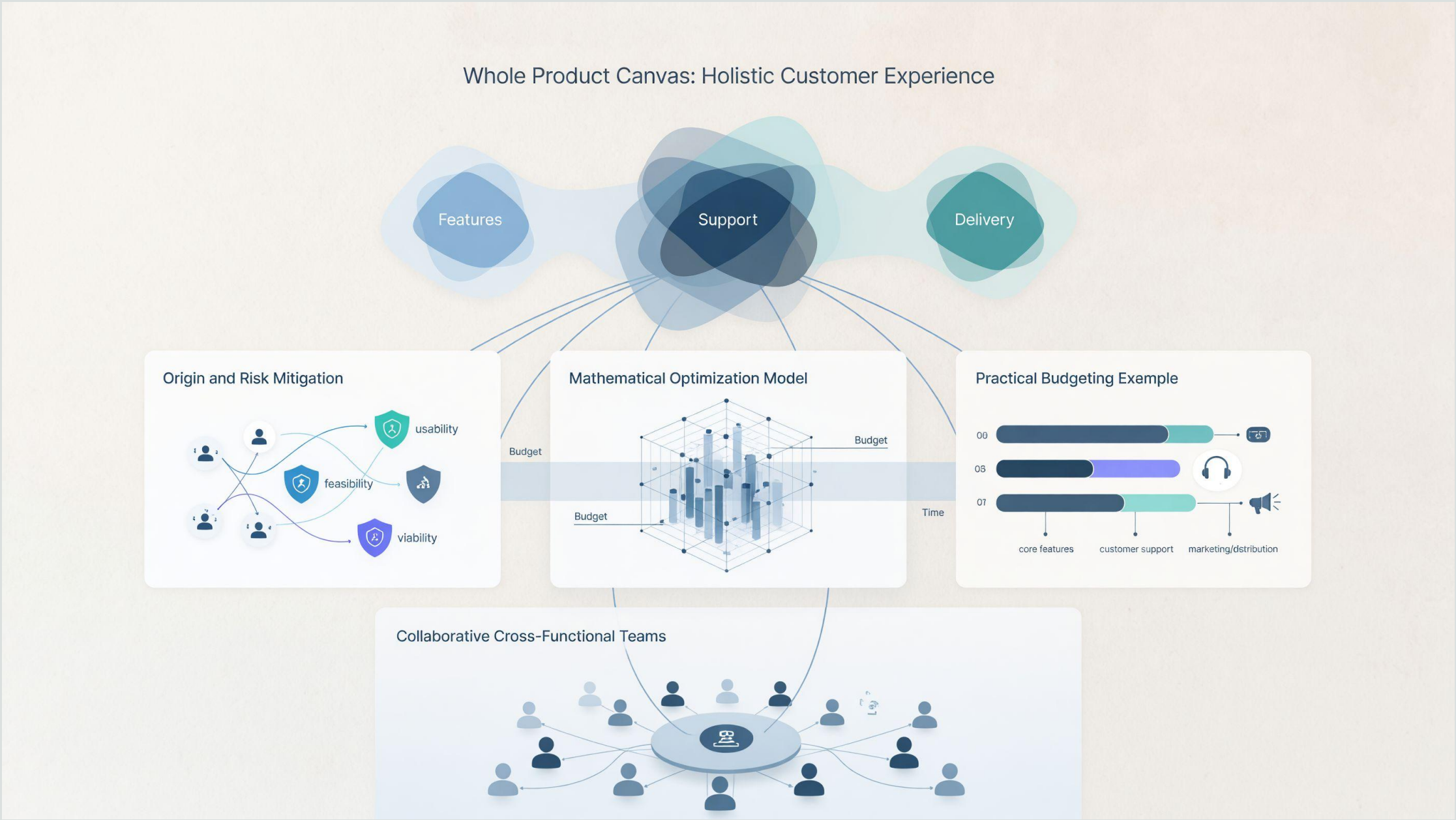
Chapter 1: Whole Product Canvas

Ir Prof Alan Lam, JP

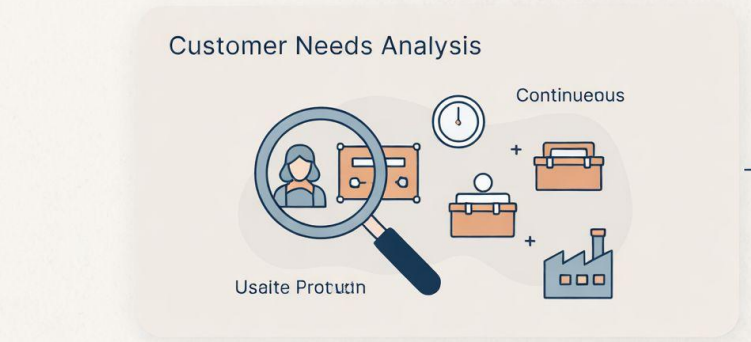


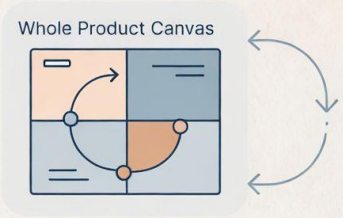
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Whole Product Canvas Framework Definition

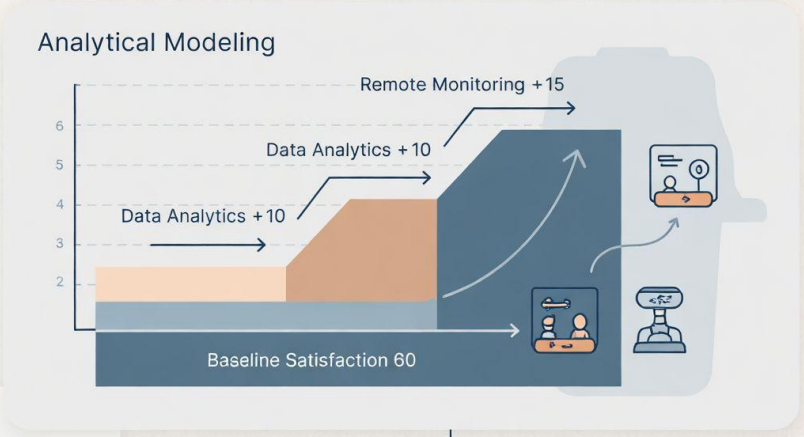


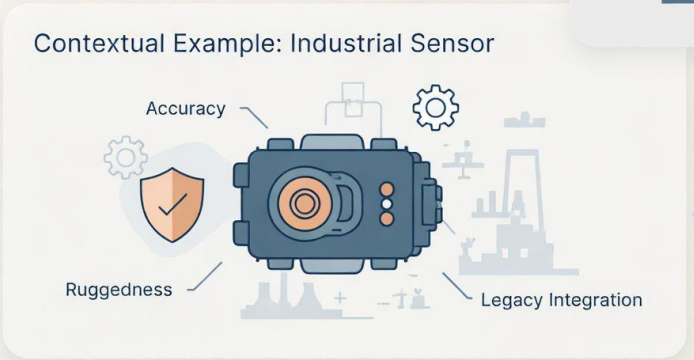
Customer Needs Identification







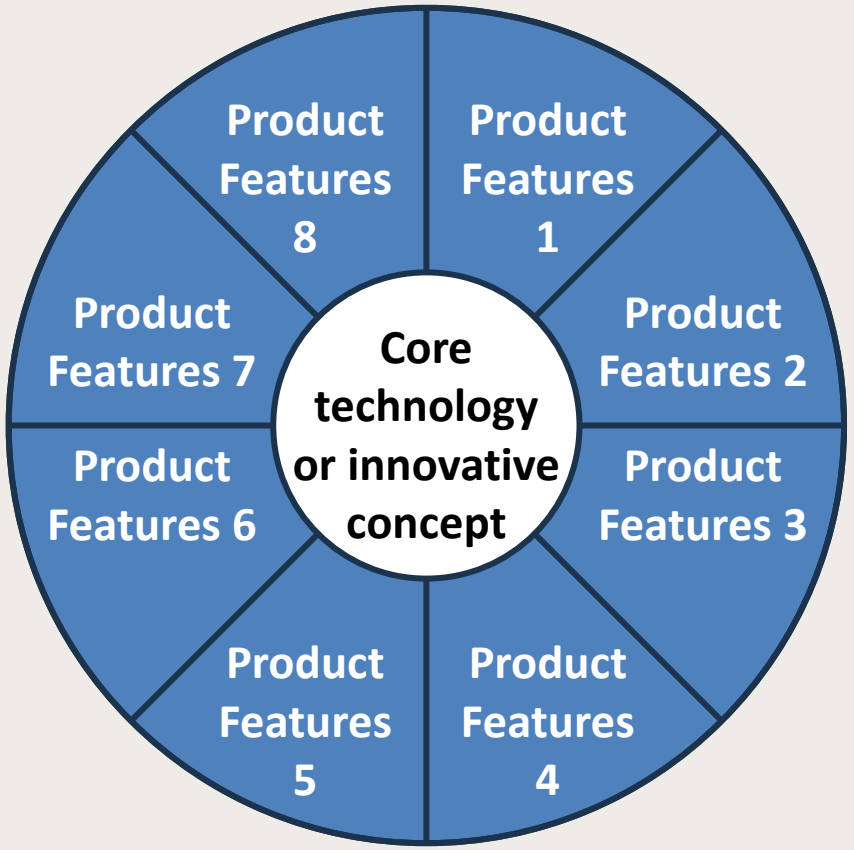






Whole Product Canvas

- For a specific target customer group with a strong reason to buy, the Whole Product is: "the full set of product and service features required to meet the customer segment's reason for buying" (Jeffrey Moore, "Crossing the Divide")
- Different customer segments have different Whole Product requirements.
- Due to the high cost of developing the entire product, the key challenge is to identify the right target customer segment and define the appropriate Whole Product (product feature set) required for that target customer group
- Product/Market Fit.
- Note: The Whole is not a "all" optional feature, but a "whole" and "complete product" composed of the required features.

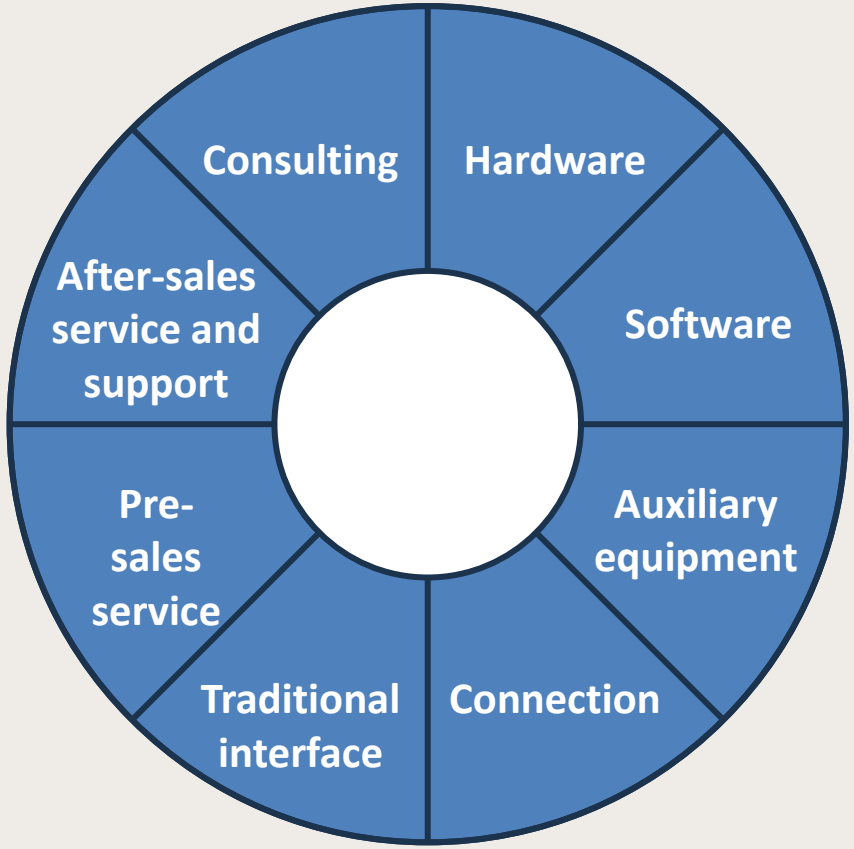


Whole Product Canvas Framework Definition

Different applications of the Whole Product: the positioning of different product characteristics

- **Example: The Whole Product Diagram of an IT product**

Complementary services

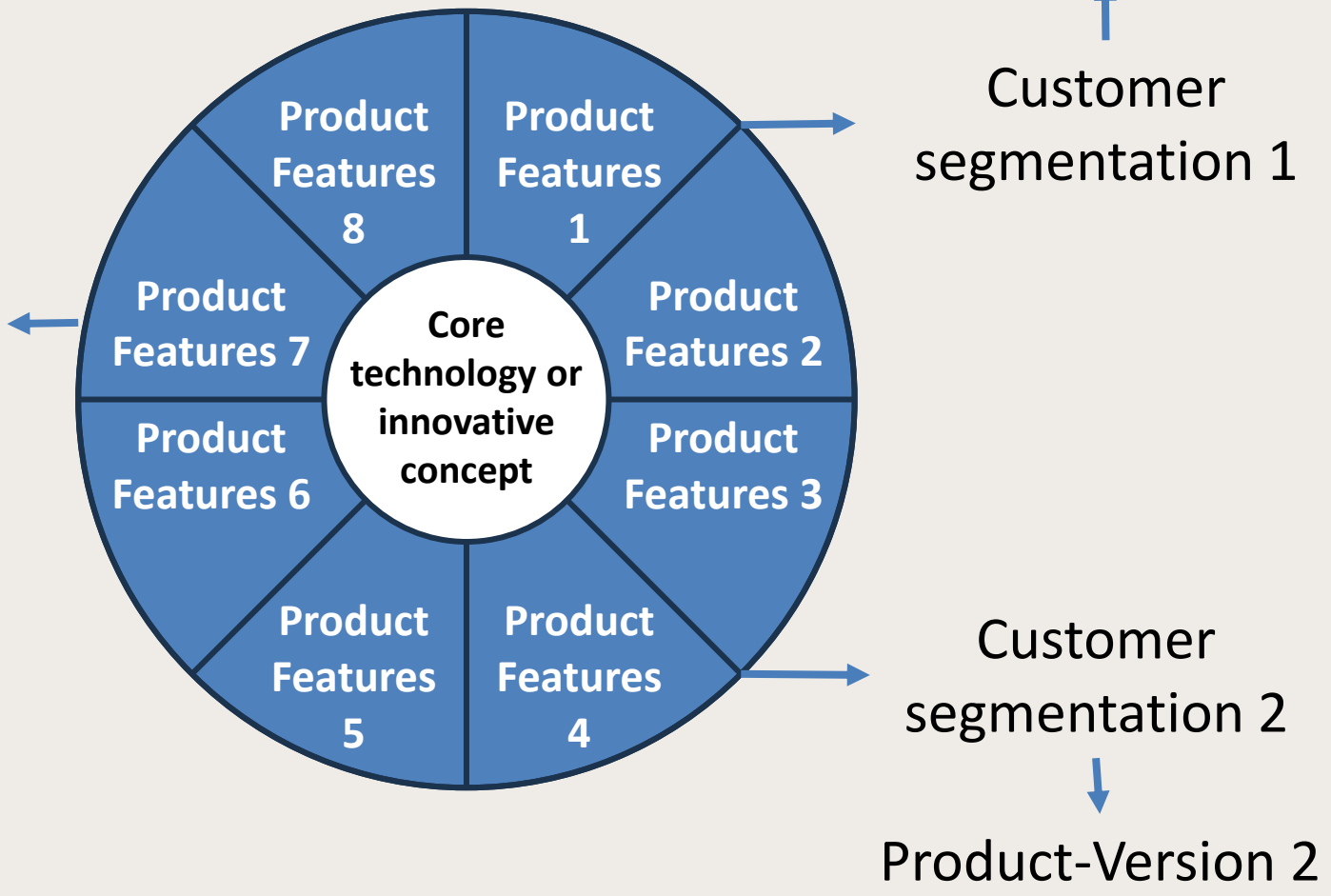


Complementary products

Whole Product Canvas Framework Definition

Different applications of the Whole Product: Build different product versions
Case: Three different product versions

- A full product is the smallest set of products and services that need to meet the purchase motivation of the target customer
- Full product map to find product-market fit points for different target markets (corresponding products (versions))

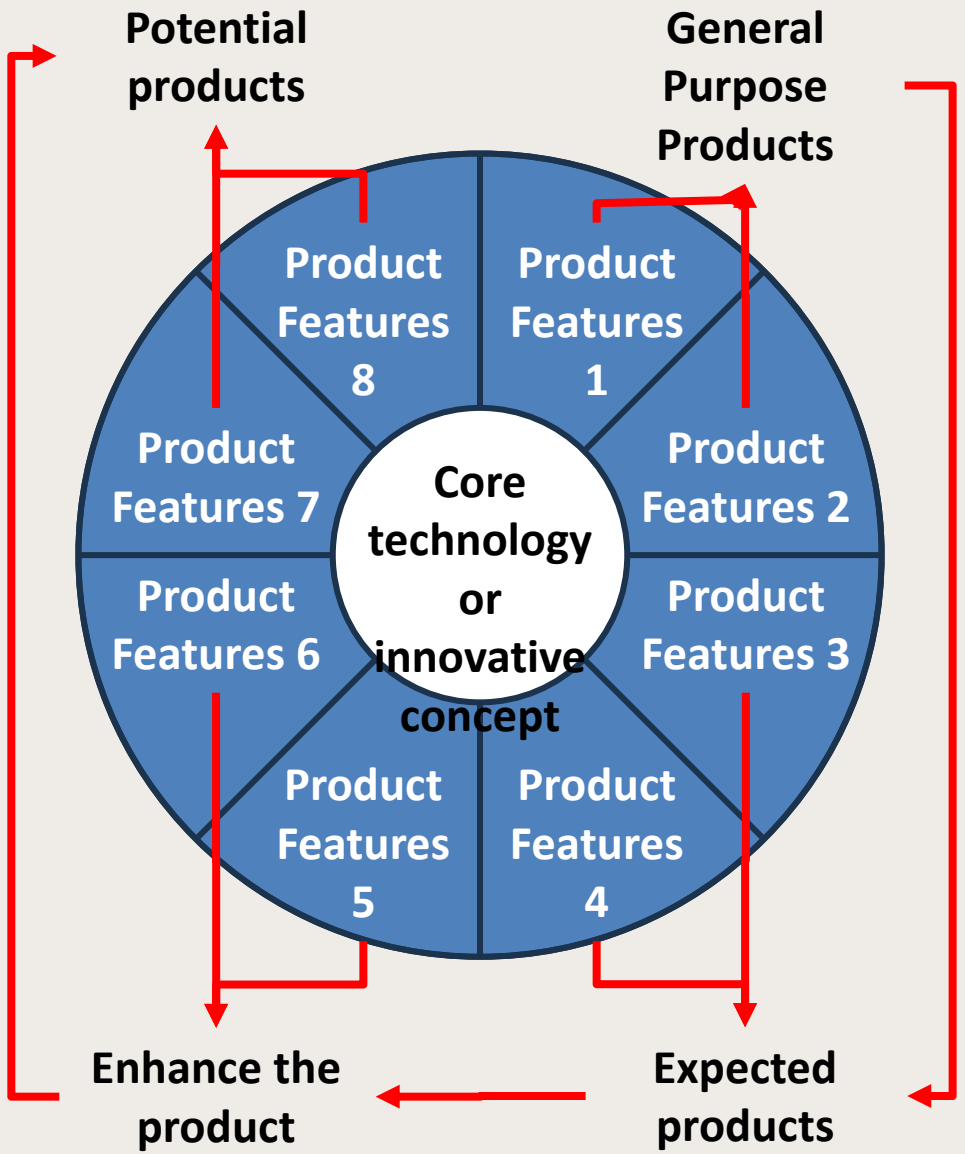


Whole Product Canvas Framework Definition

Lecture 06) Product Development

Different applications of the Whole Product Canvas: different levels of the Whole Product

- The whole product can be selected according to the core technology (product/service) and can be divided into different versions: Universal Product (Version): The core function of the product; It meets the most basic needs in the product category. For example, for browsers, the core function is web browsing.
- Intended Product (Version): The functionality and features that customers expect to receive when purchasing the product; Including the basic availability, reliability, performance, etc. of the product. For example, the browser should support multi-platform (iOS, Android, Windows) and other web standards.
- Enhanced products (releases): Those that exceed basic customer expectations and add value to improve customer experience and satisfaction. This may include training, customer support, customized services, third-party integrations, warranties, etc.
- Potential products (versions): the possible future development direction of the product, features and services that have not yet been realized but have potential value; It represents the long-term development path and innovation space of the product.



Value Proposition Mapping

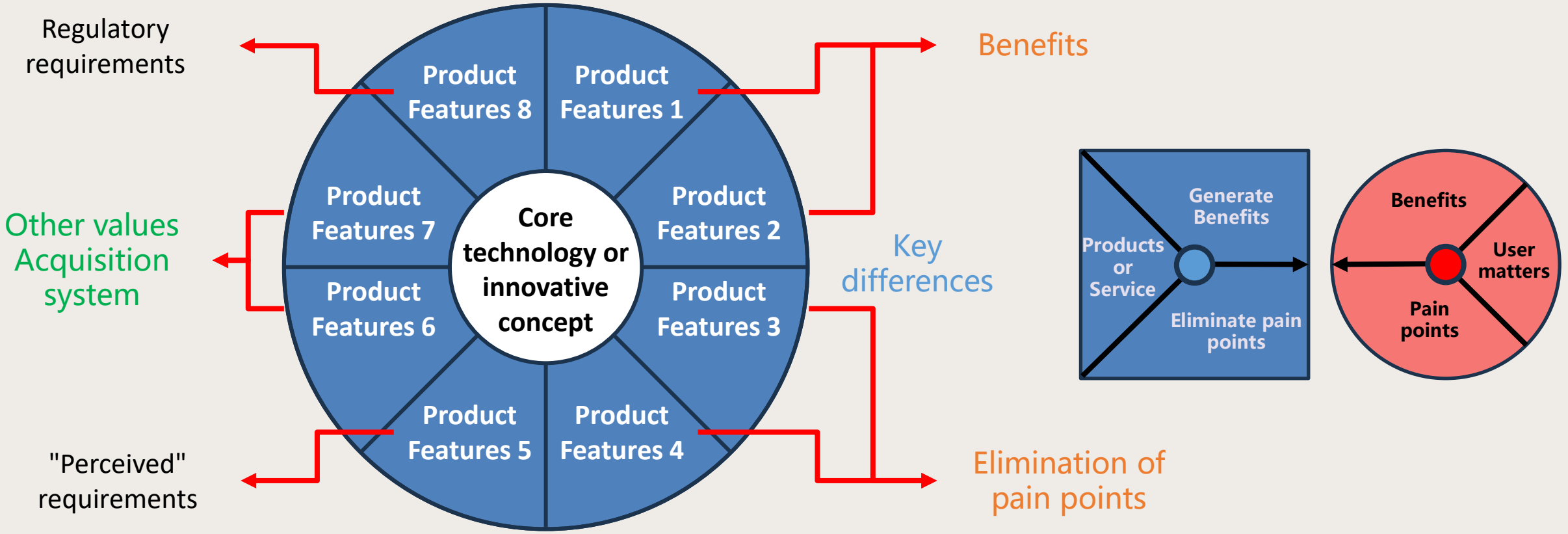


Whole Product Canvas Framework Definition

Lecture 06) Product Development

Build a product value organization based on different product characteristics

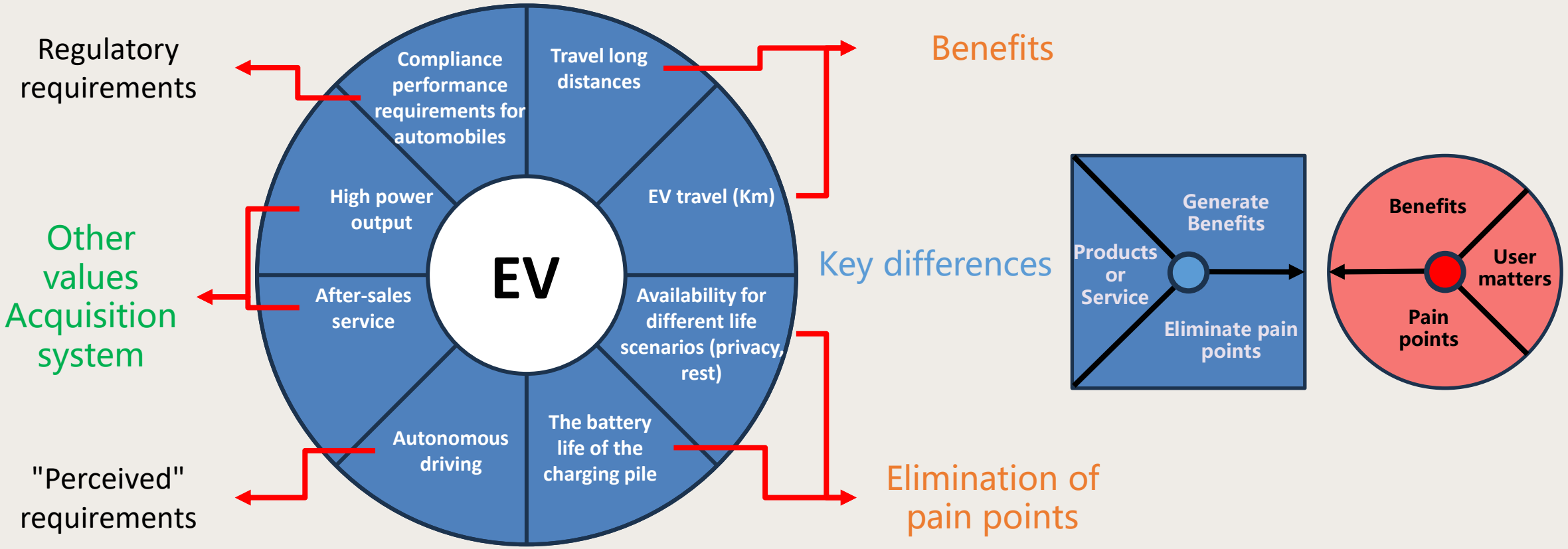
- **Case Study: Whole Product Canvas (WMP) vs. Value Proposition Map (VPC)**



Whole Product Canvas Framework Definition

Lecture 06) Product Development

Use the left half of the technology (project) VPC to draw the Whole Product Canvas
Case study: Whole Product Canvas (WMP) and value proposition map (VPC) of electric vehicles

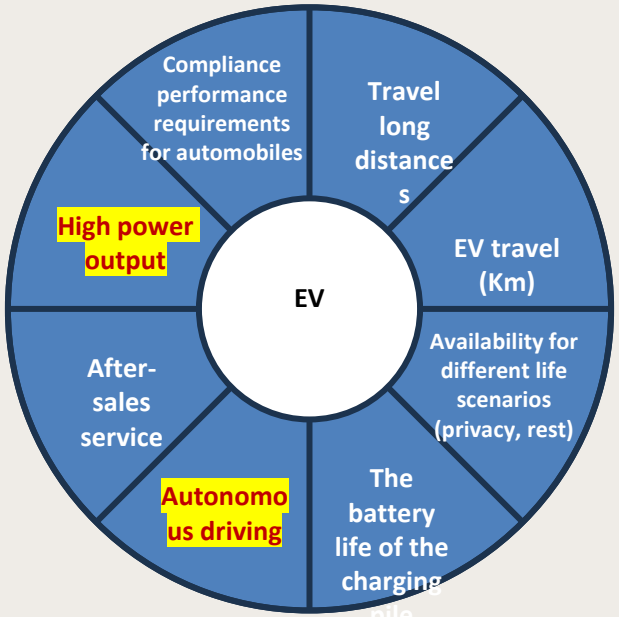


Whole Product Canvas Framework Definition

Build multiple versions of your product by selecting a combination of features in the Whole Product Canvas

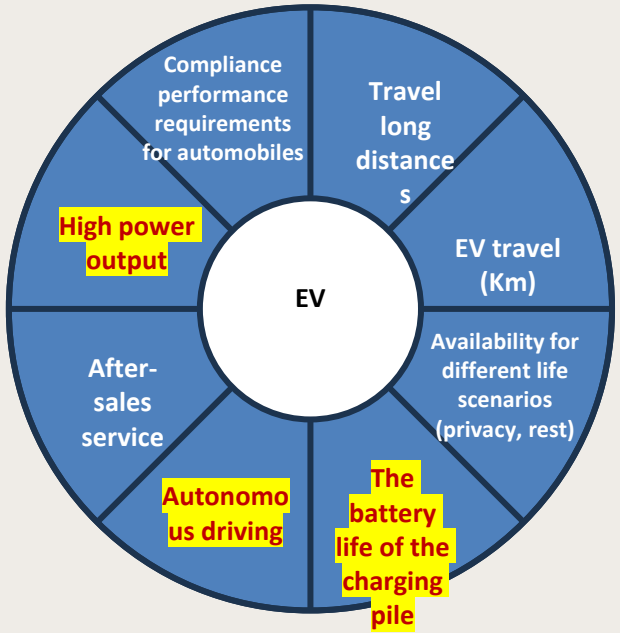
- Case: Whole Product Canvas of electric vehicles

Electric vehicles are currently common



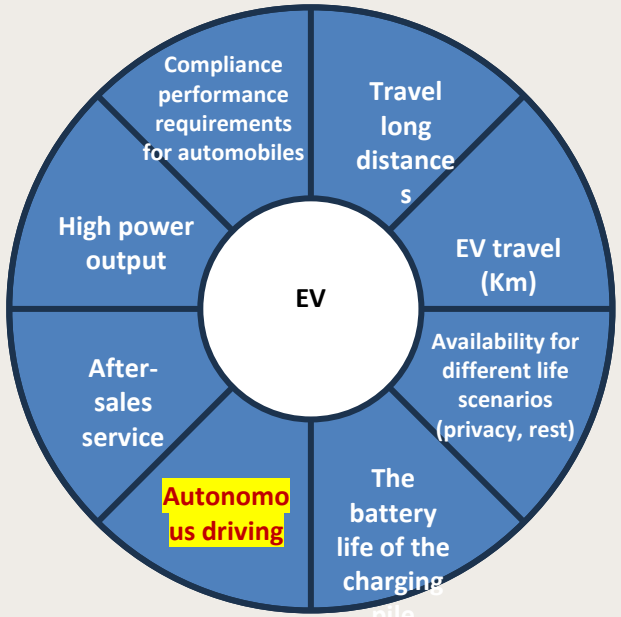
General Purpose Products

Electric vehicle case of battery swapping: NIO



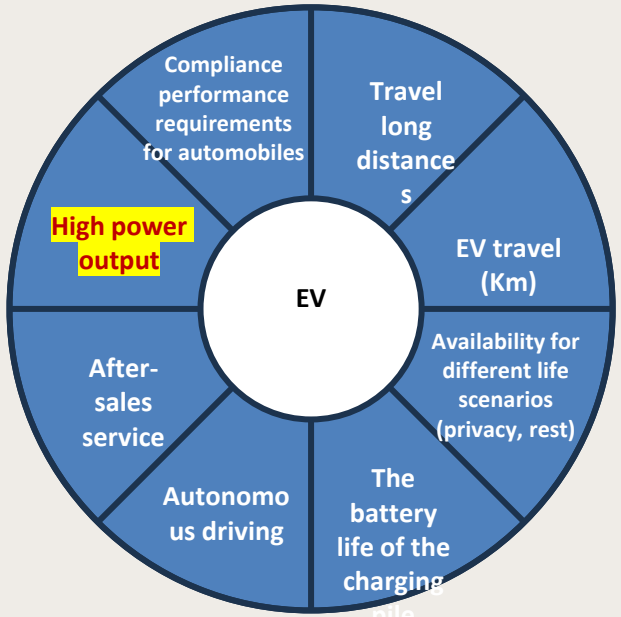
Expected products

High-performance electric sports car example: Tesla Roaster SU7 Ultra



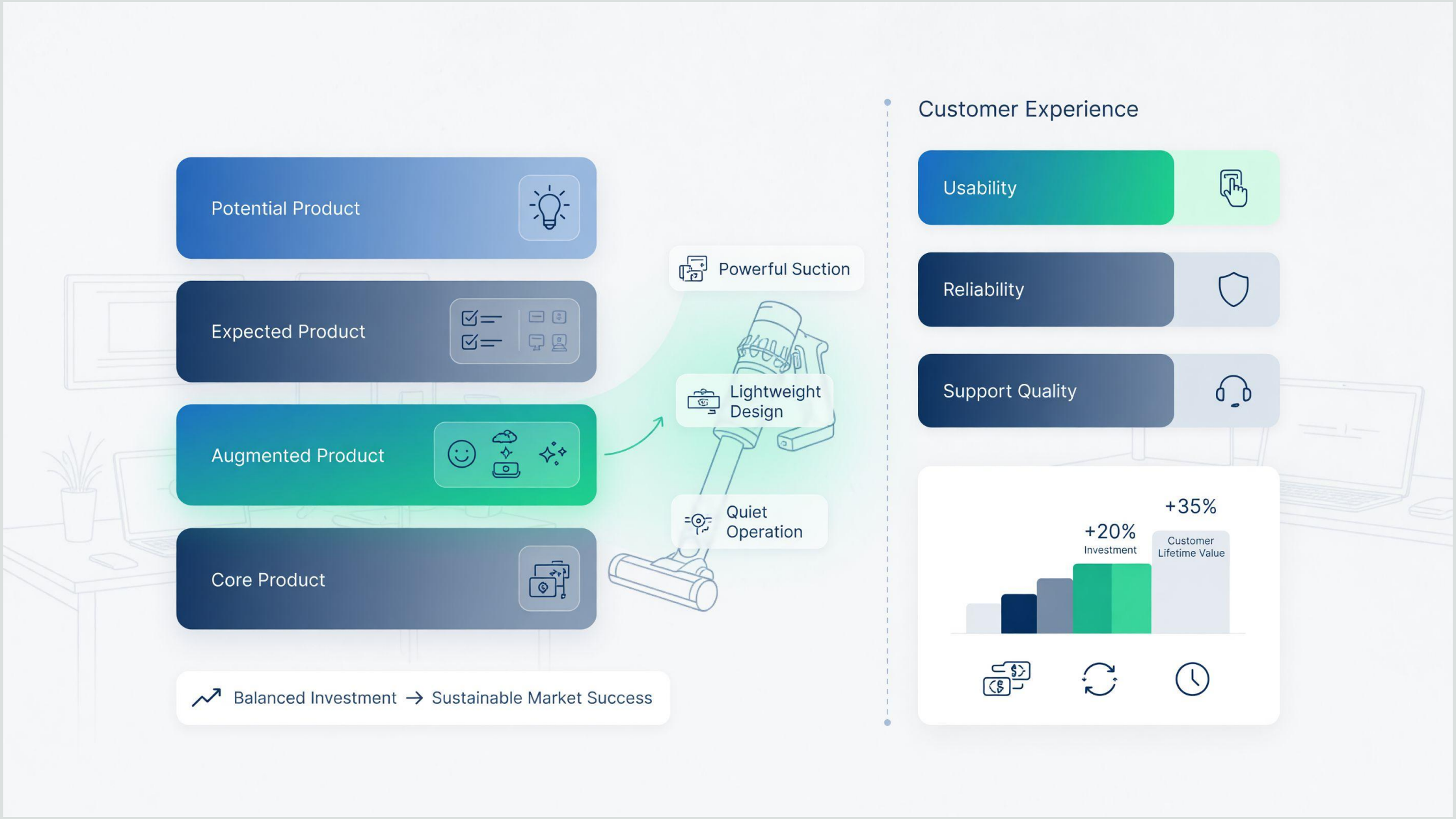
Enhance the product

Fully automatic electric vehicles, Electric Vehicles for the Elderly (Future) Unmanned taxis, driverless cars, long-distance hotel bedding? Mobile karaoke hall?

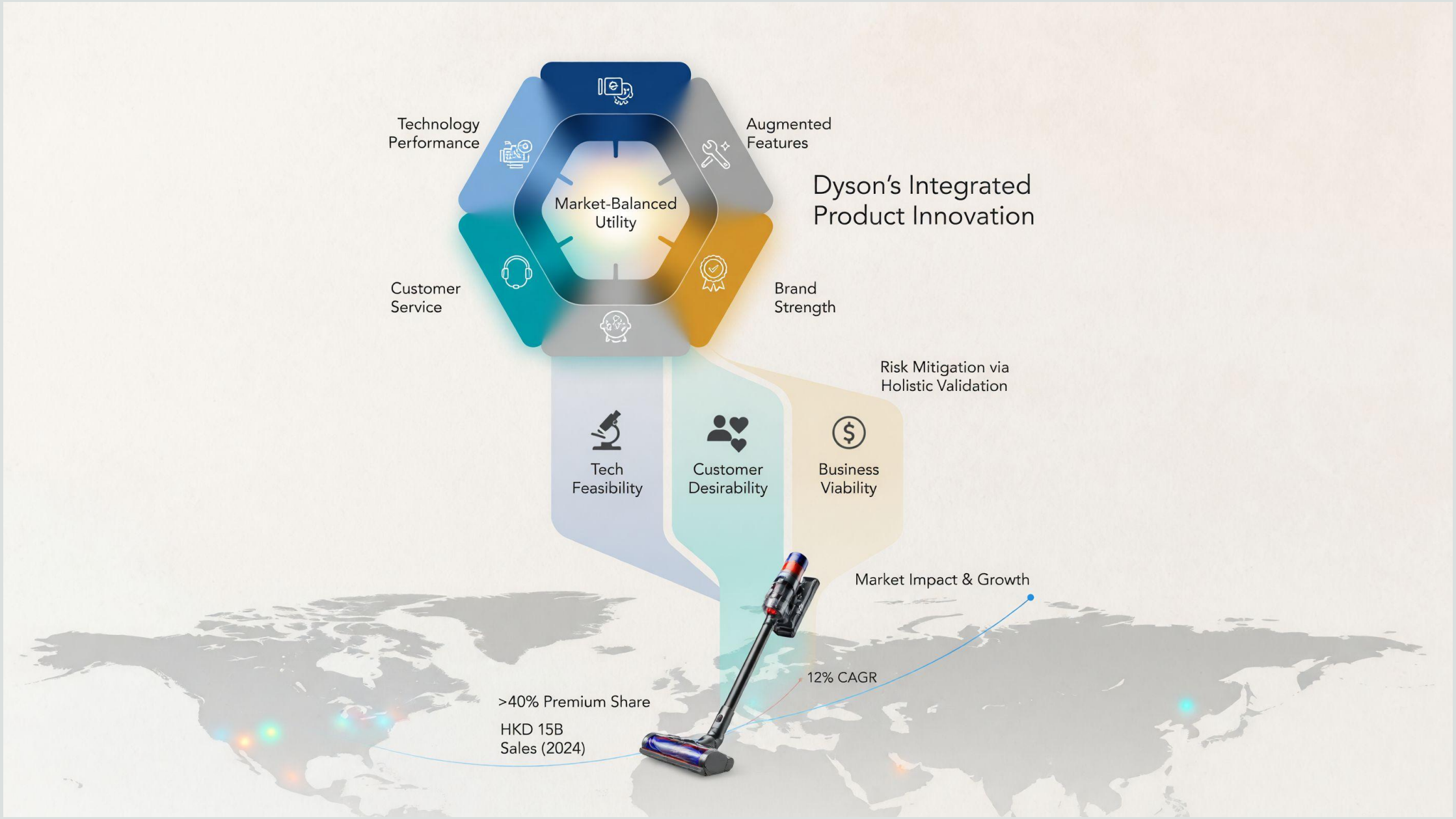


potential products

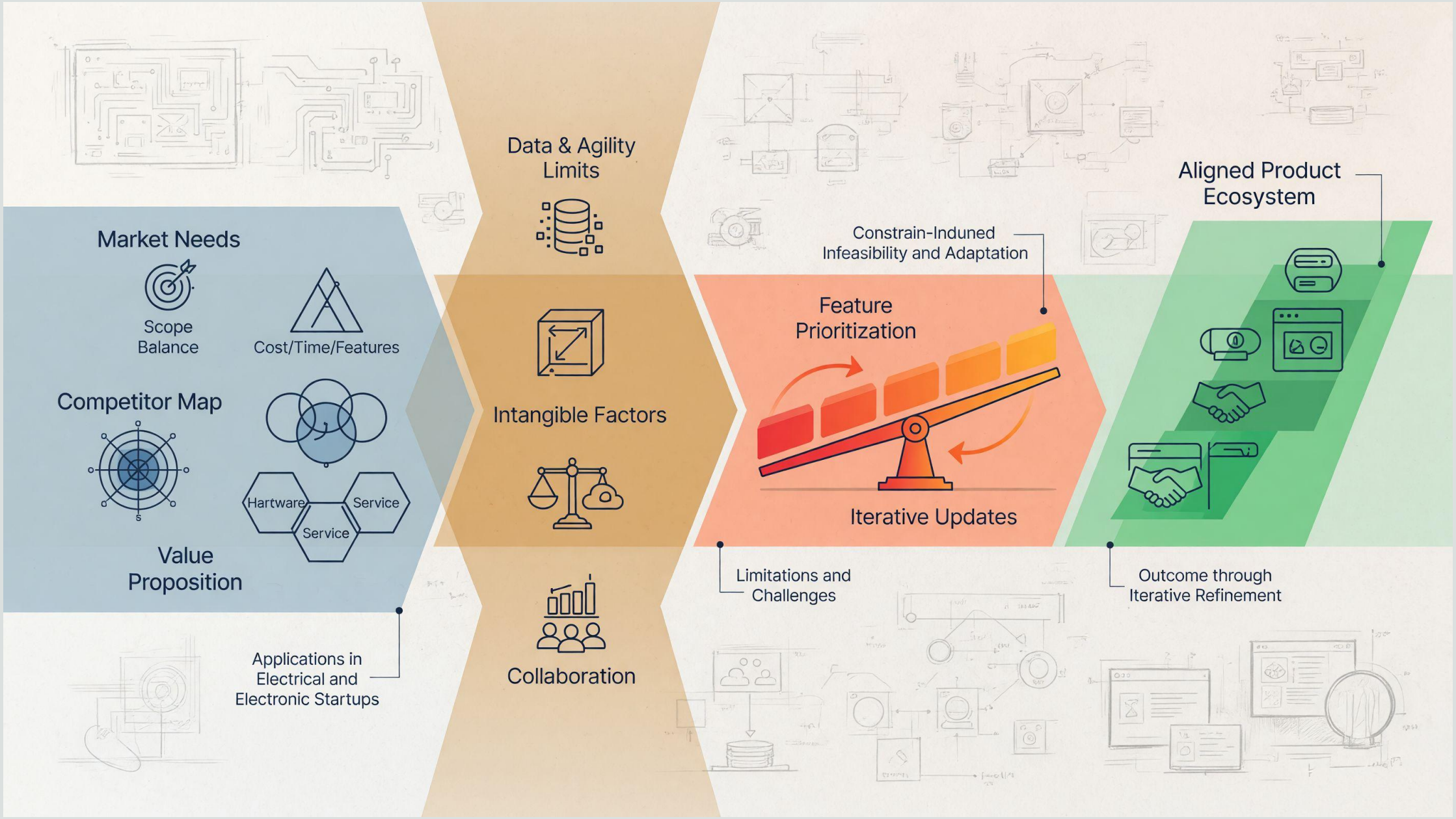
Whole Product Levels and Customer Experience



Case Study of Technology Products using Whole Product Canvas



Applications and Limitations



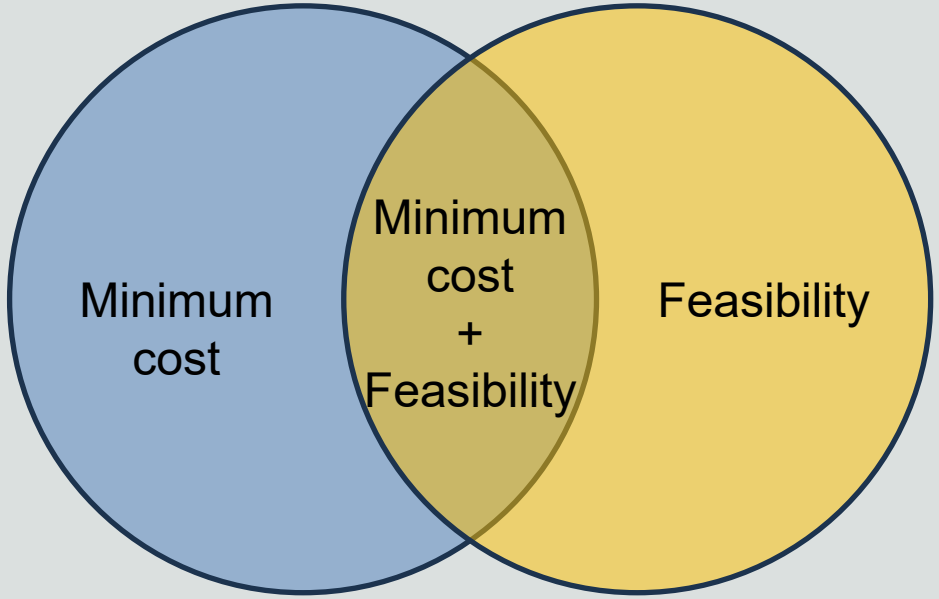
Chapter 2: Minimum Viable Product

Ir Prof Alan Lam, JP



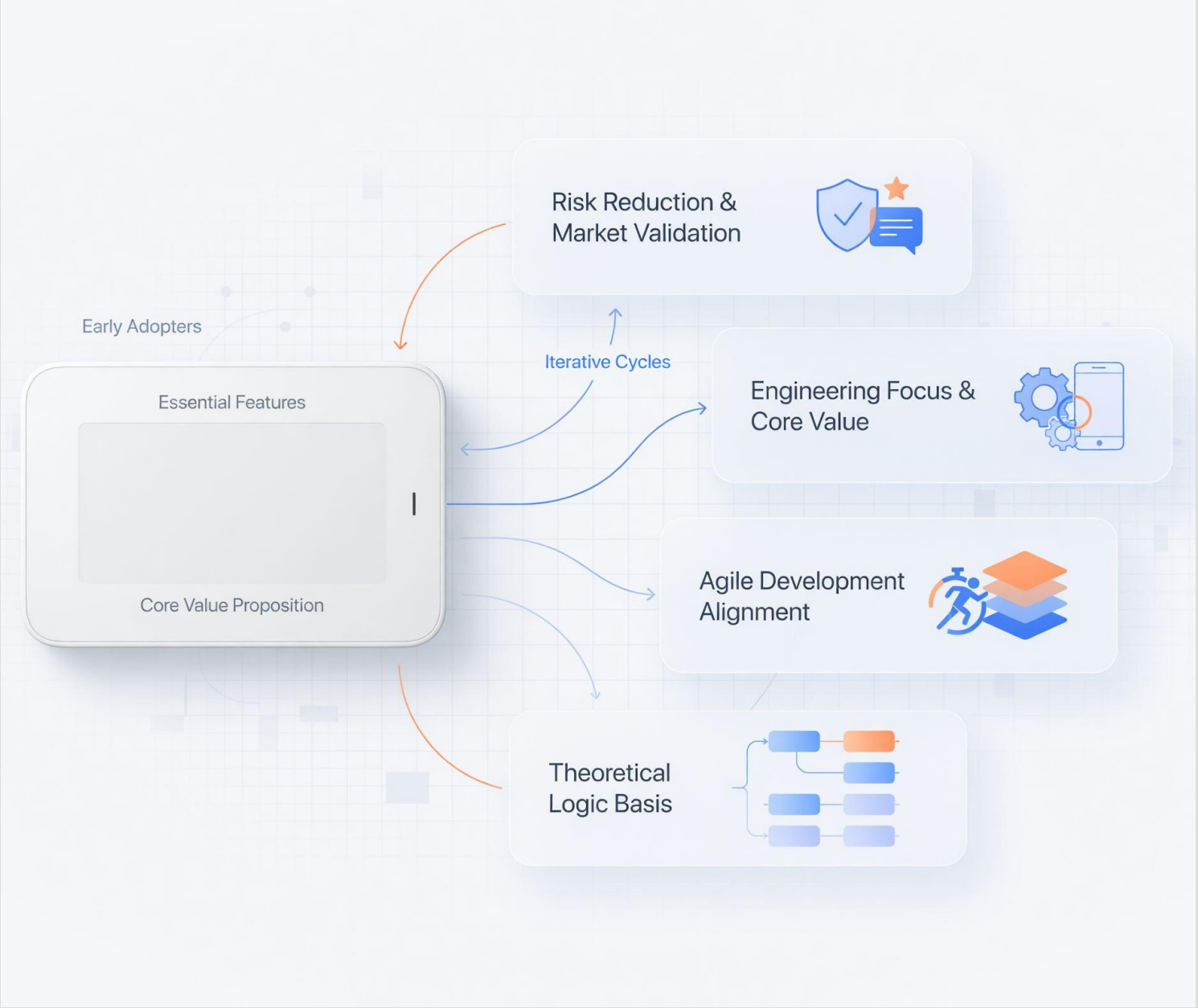
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Definition of Minimum Viable Product (MVP)



- "A minimum viable product (MVP) is a version of a new product that allows teams to gather the maximum amount of validation learning about customers with minimal effort. "
- "An MVP is not just a minimal product, it is a strategy and process designed to sell a product to customers. It is an iterative process that includes idea generation, prototyping, presentation, data collection, analysis, and learning."

– Eric Reese
(Author of Lean Startup)

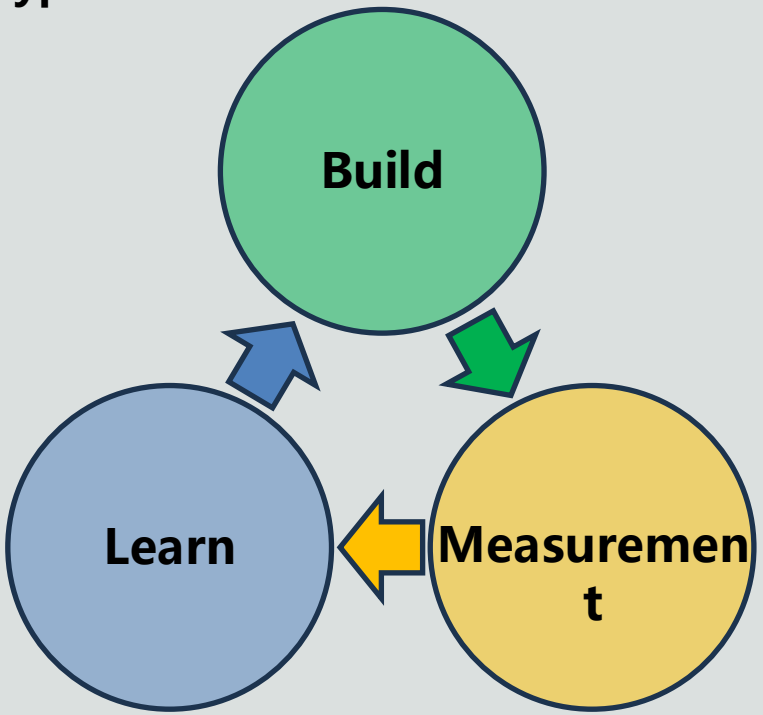


MVP Feedback Loop:

- **The Minimum Viable Product (MVP) feedback loop is a core component of the Lean Startup methodology.**

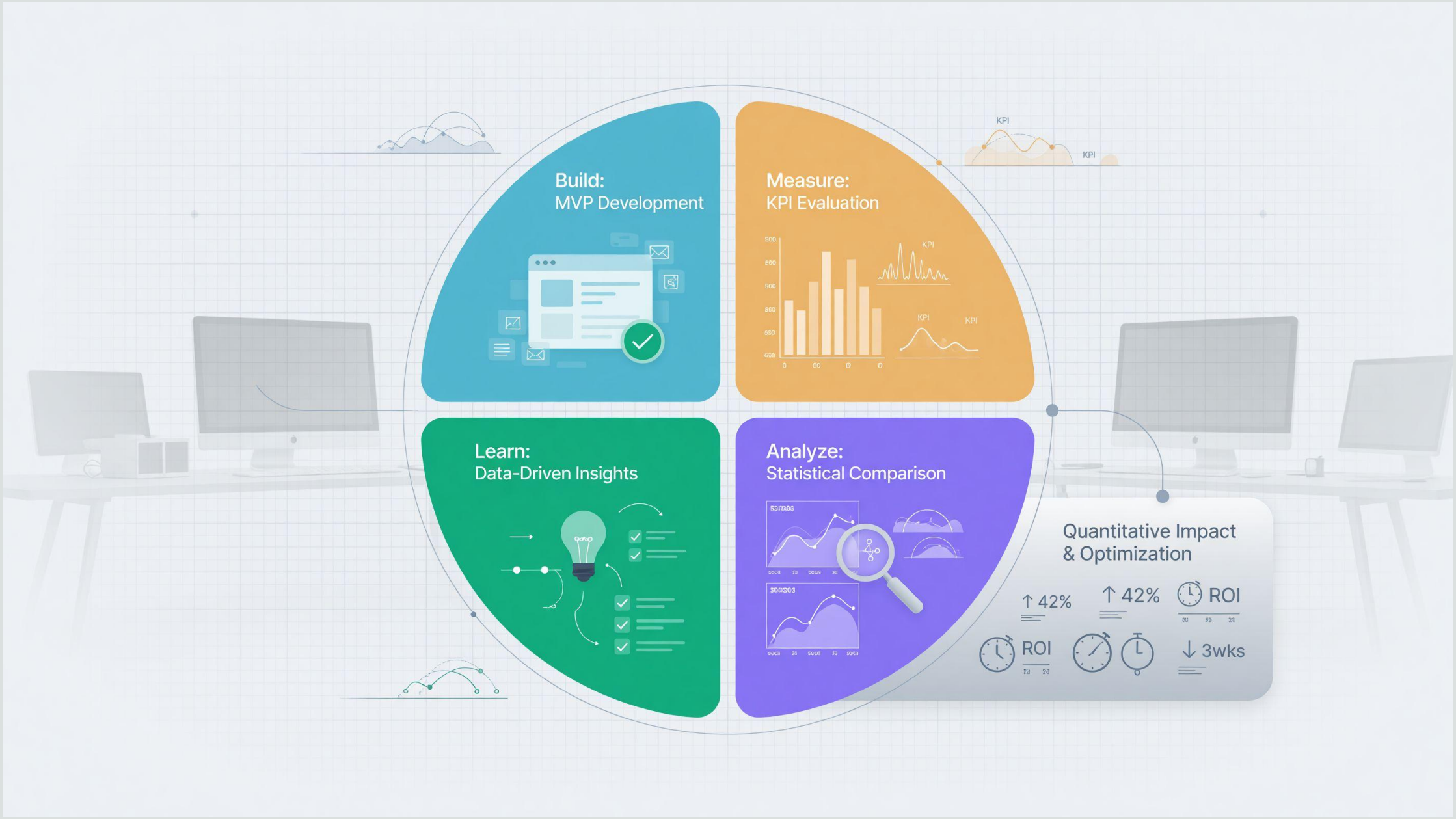
Build:
Identify market solutions that require acceptance and develop minimum viable products (MVPs) to start the learning process as quickly as possible. The MVP should contain only enough functionality to test the core theory or hypothesis and be able to collect validation learning.

Study:
The collected data and feedback are analyzed to validate or refute the initial hypothesis. Decide whether to "transform" (significantly change direction) or "persist" (including iterative improvement).



Measurements:
Put into use at MVP to collect data on its performance in the market, focusing on actionable metrics that can guide business strategy. Including customer feedback, direct interactions, emotional responses, and more.

Iterative MVP Development Cycles

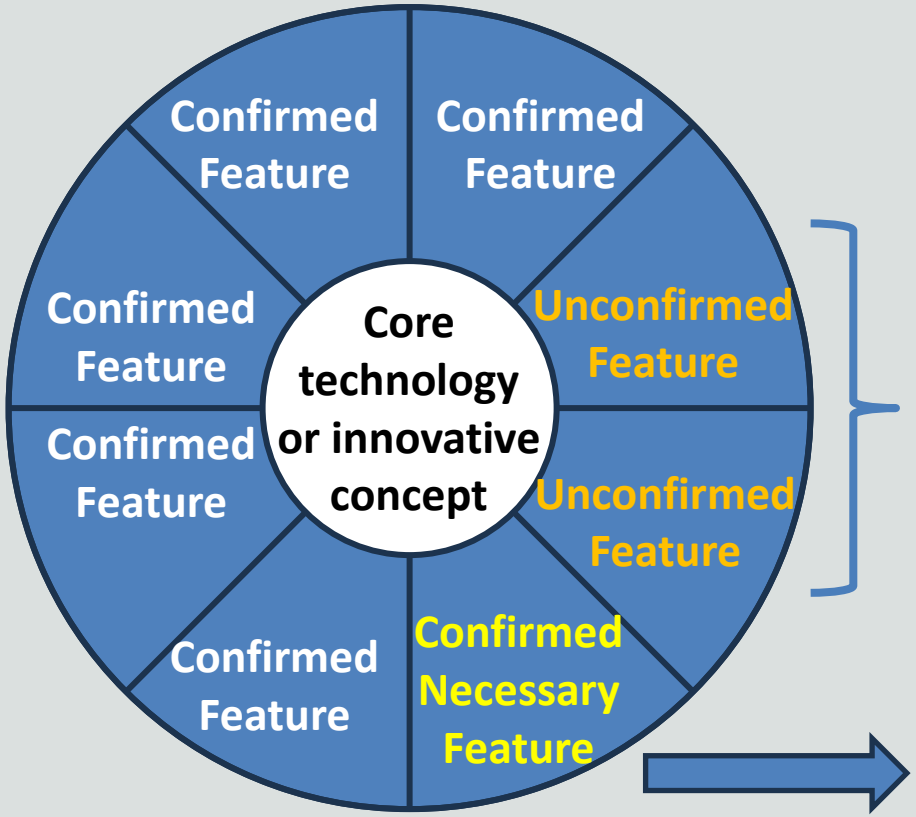


Example of Risk Assessment Models using MVP Data



Minimum Viable Product (MVP) vs. Whole Product (WPC):

- **With WPC, select unverified product features: We assume that customers want new product features that have not yet been validated.**
- **First, determine the list of all potential characteristics (WPC) of the product (achievement, project);**
- **This list is then strictly filtered, leaving only the MVP "minimum" and "must-have" features.**
- **Add new features to older products: Validate the assumption that the product has new features that customers want but have not yet validated (not the original required features) with the minimum viable product.**



It is assumed that feature A is the basis of the basic and necessary functions (guaranteed to be feasible) of this type of product

The core of MVP (Verification Point) - "Minimum"

Minimum viable product

Chapter 3: Scope of Work

Ir Prof Alan Lam, JP

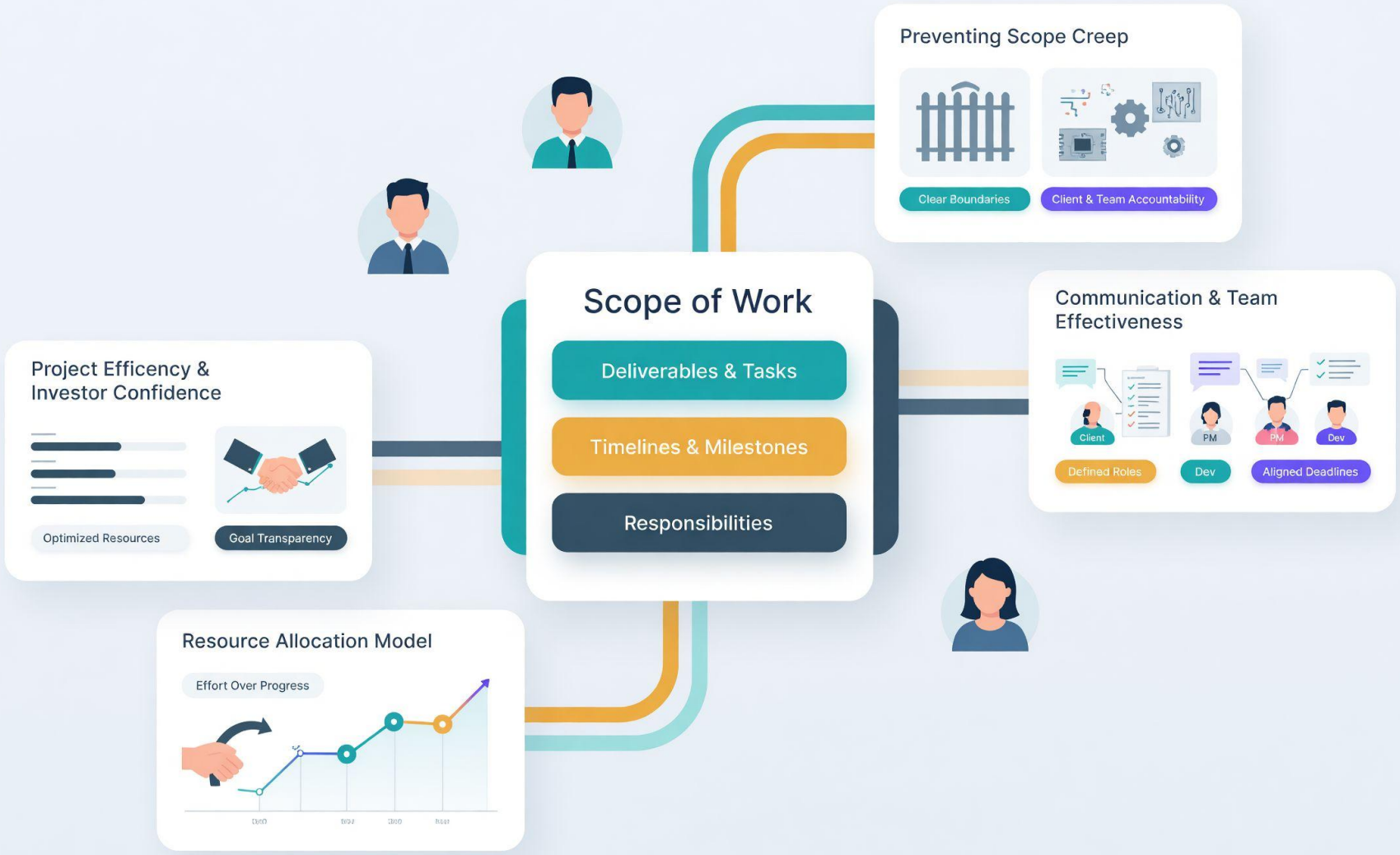


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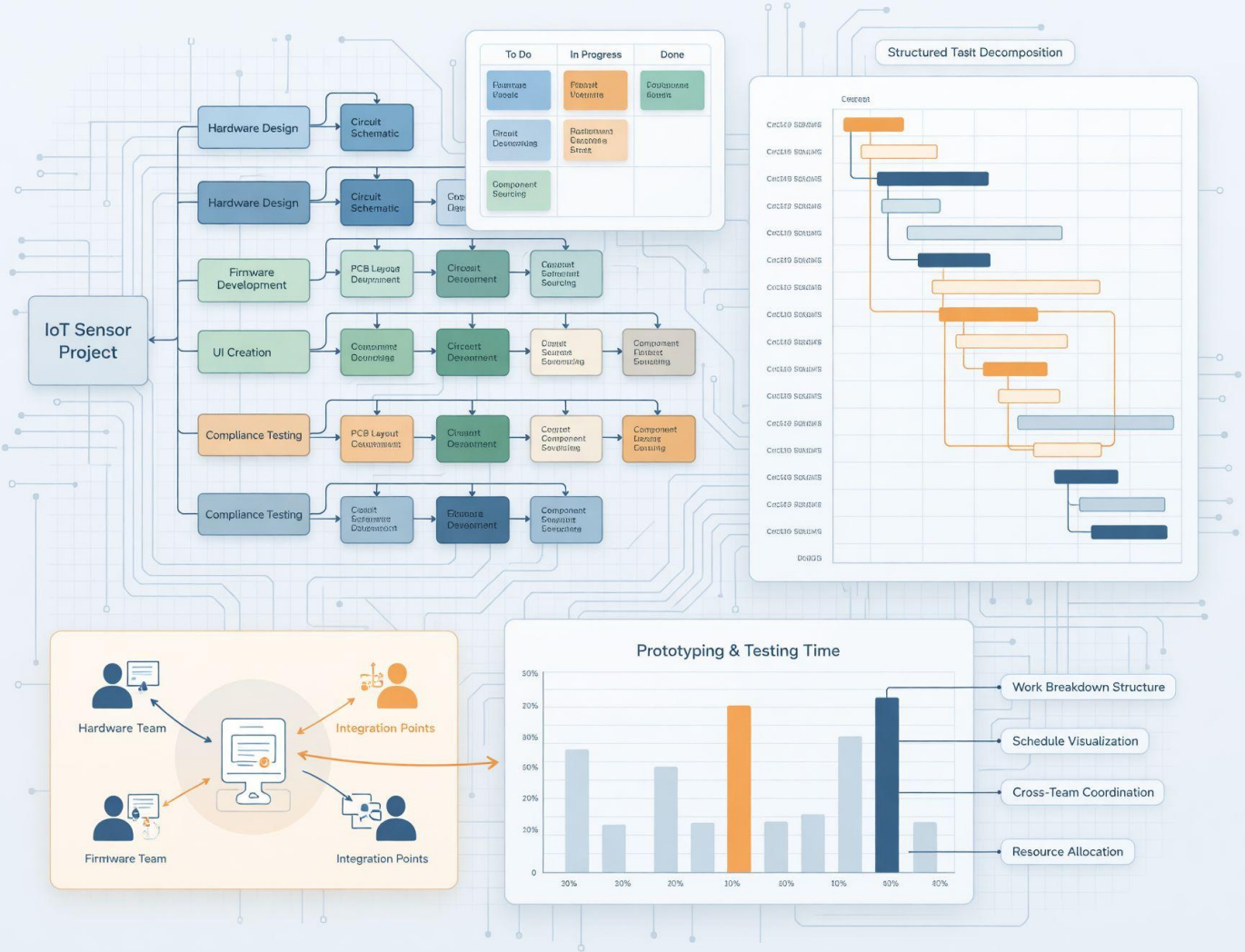
Introduction to Project Scope



Scope of Work Definition and Importance



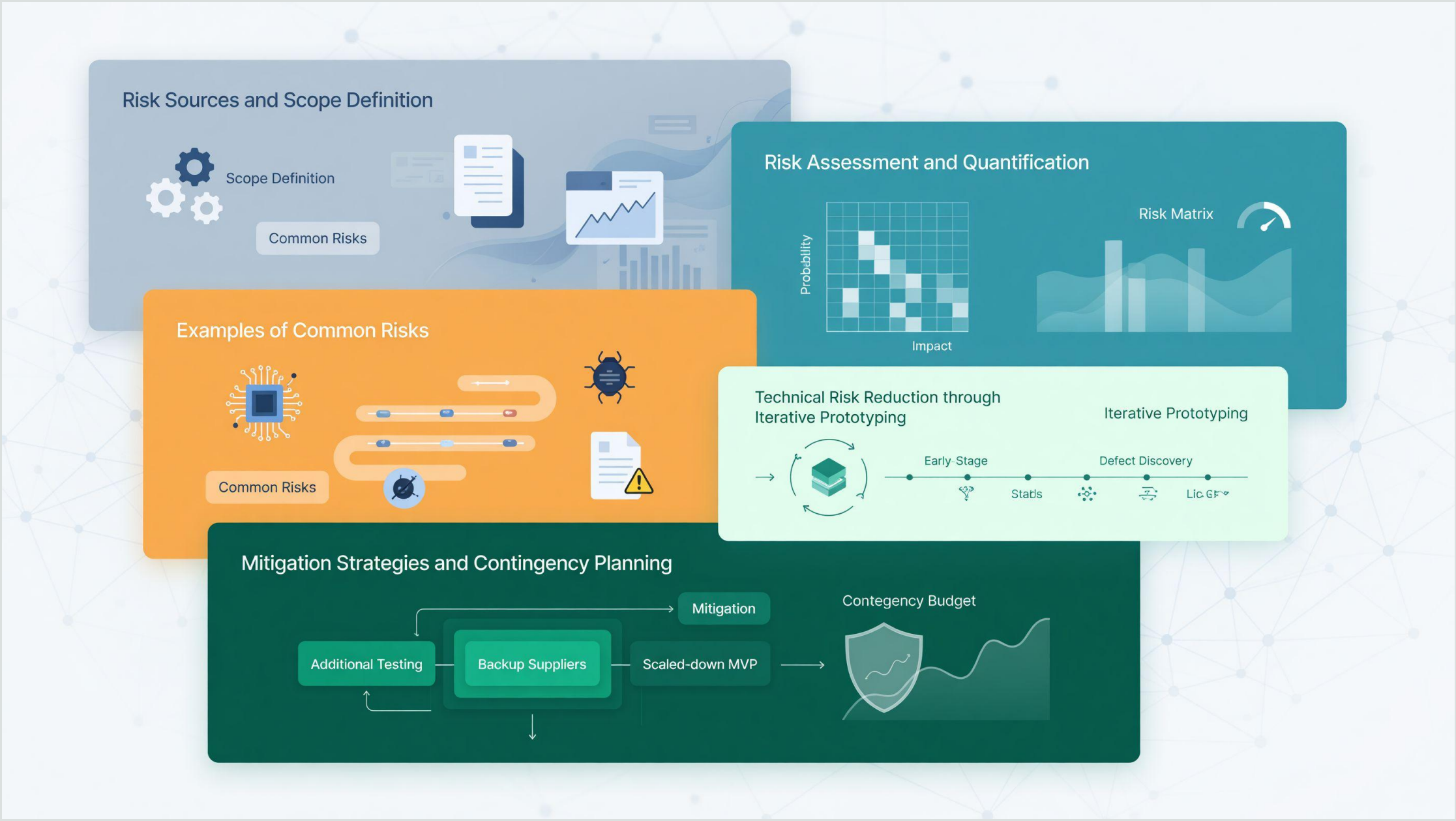
Task Breakdown for Technology Startups



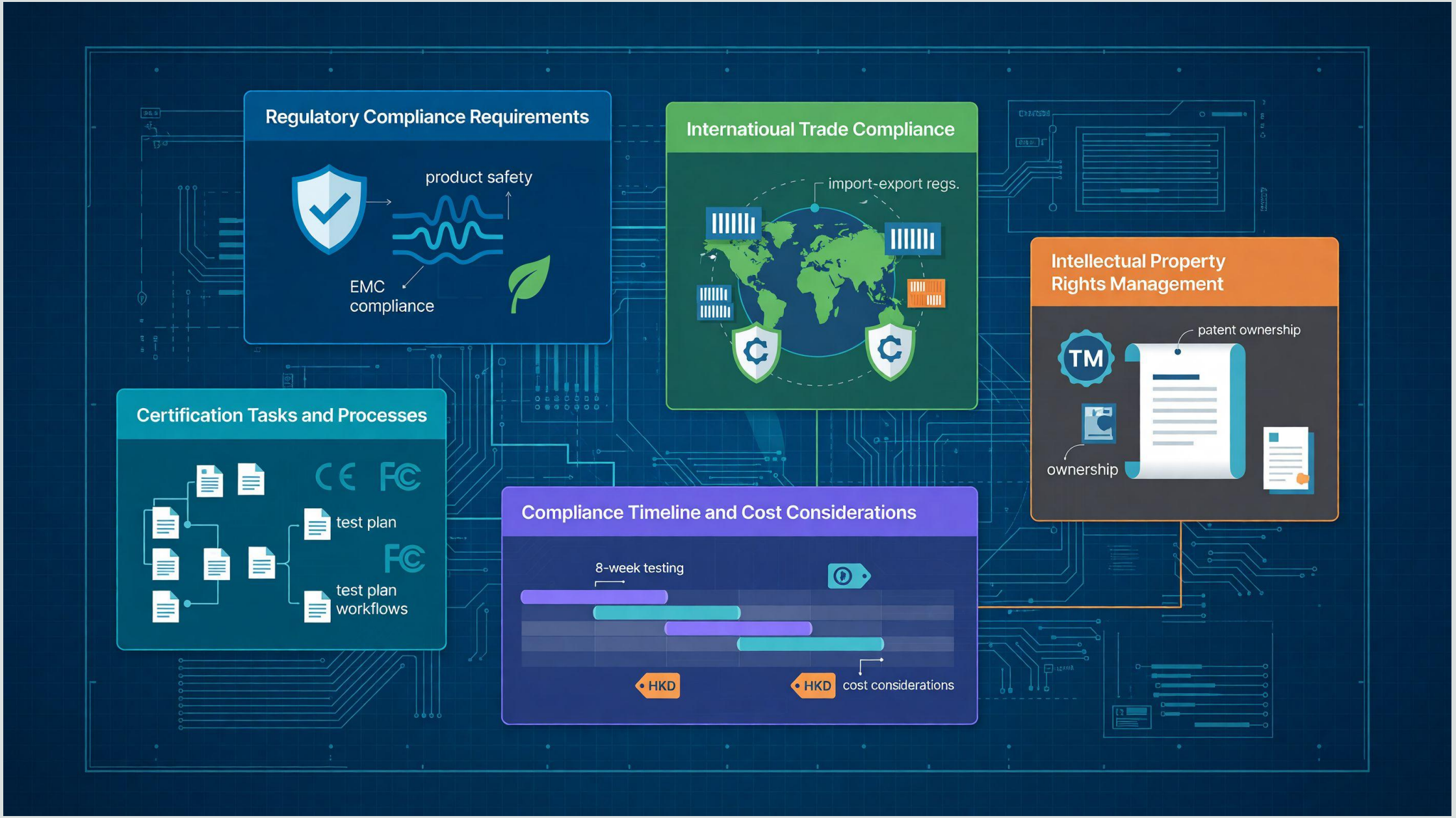
Resource Allocation within Scope



Risk Identification and Mitigation Strategies



Legal and Compliance Aspects in Scope of Work



Managing Scope Changes

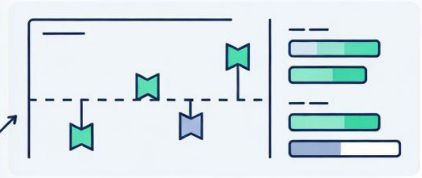


Scope Documentation Best Practices

Standardized Documentation Framework

- Objectives
- Deliverables
- Constraints
- Assumptions
- Acceptance Criteria
- Responsibilities

Schedule Integration and Version Control

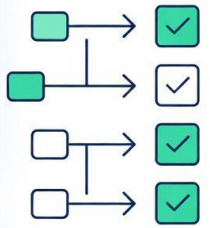


Milestones & Version Control

Clear and Accessible Language



Clear, Jargon-Free Language + Visual Aids



Continuous Review and Improvement



Regular Review & Lessons Learned

Impact on Project Success



Risk Reduction • Resource Optimization • Successful Launch

Chapter 4: Setting Milestones and Key Performance Indicators (KPIs)

Ir Prof Alan Lam, JP



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Introduction to Project Milestones



Setting Effective Milestones for Technology Projects



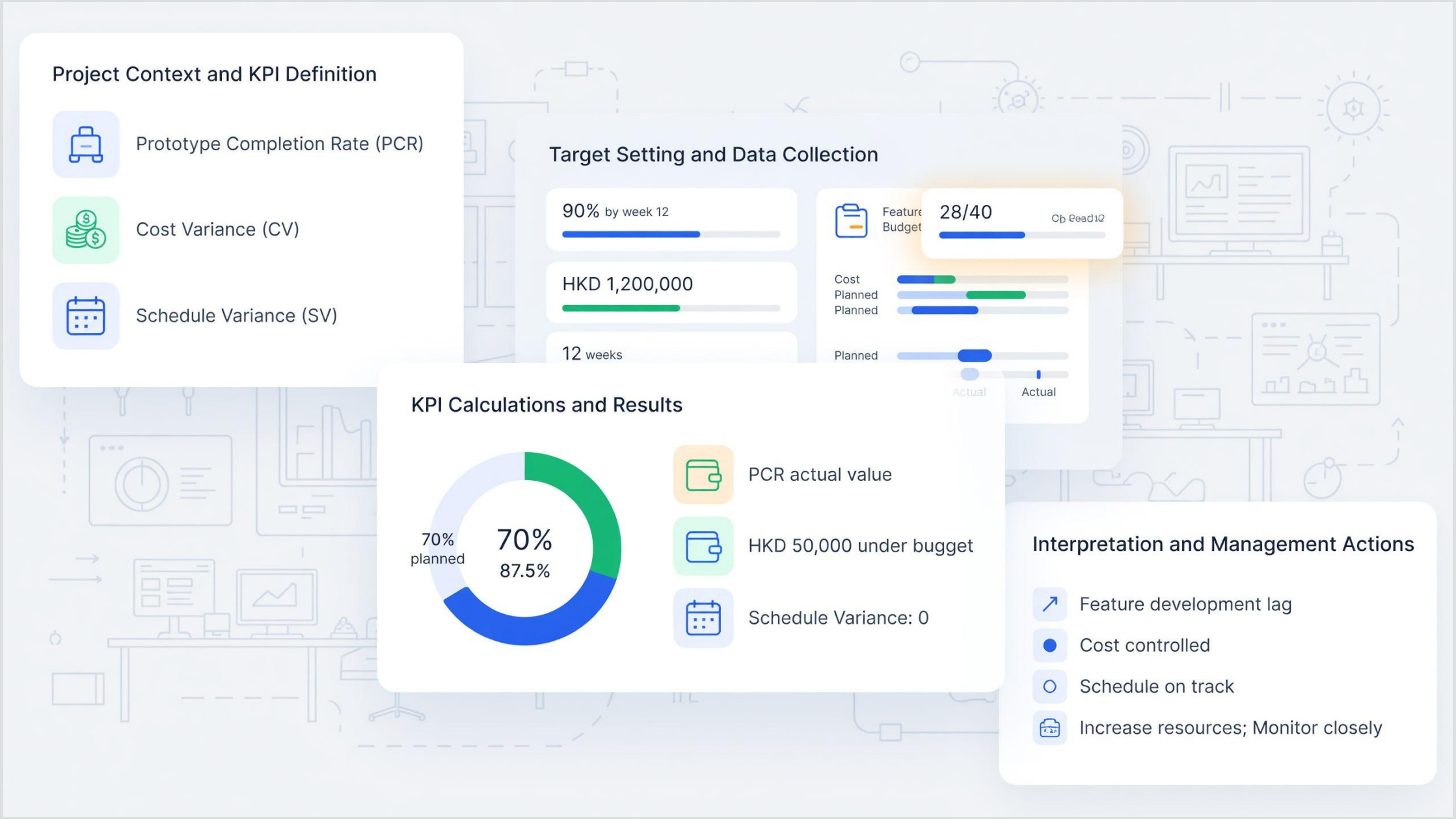
Introduction to KPI



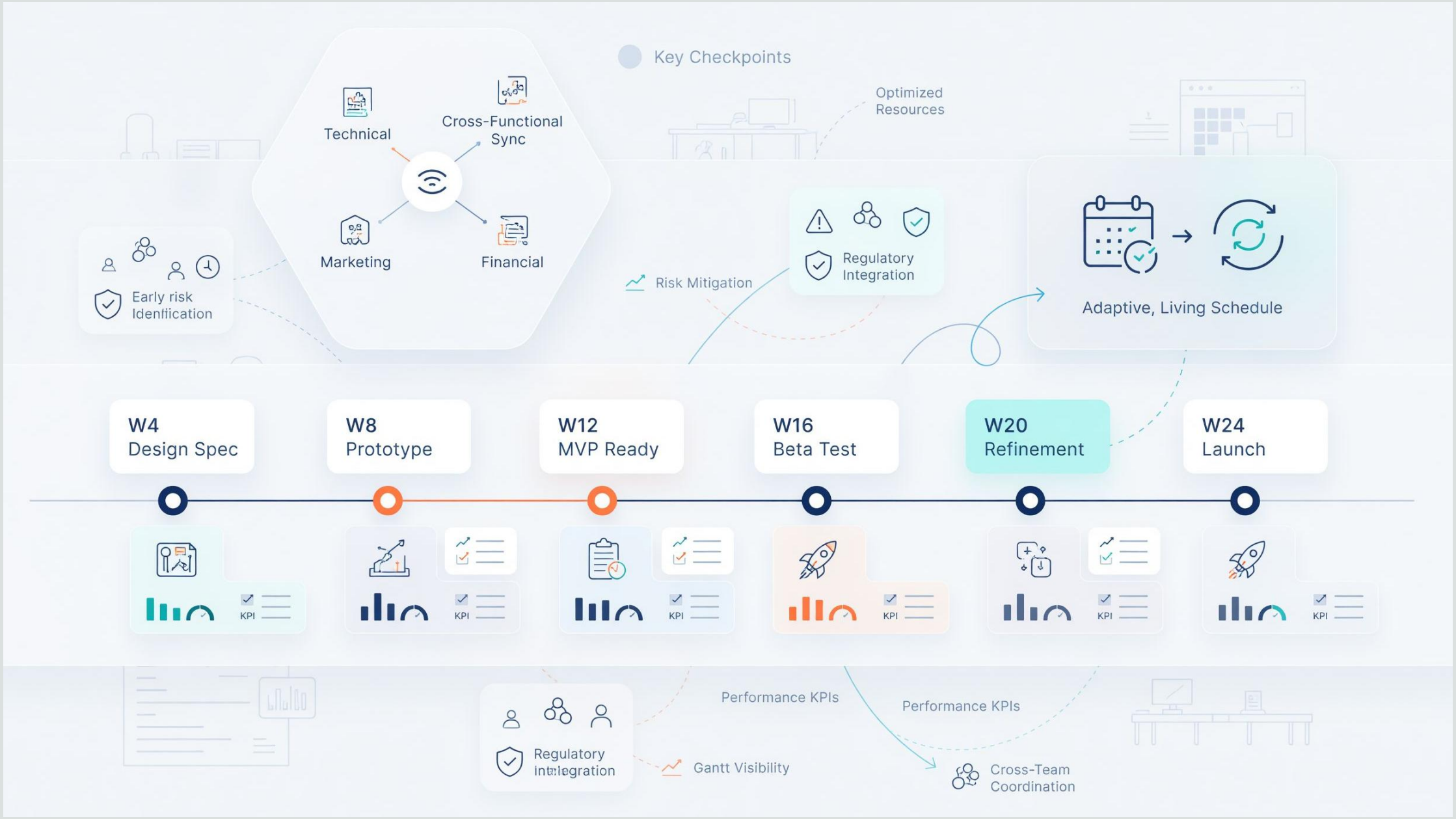
Definition and Selection of Key Performance Indicators (KPIs)



Step-by-Step Example: KPI Calculation Example



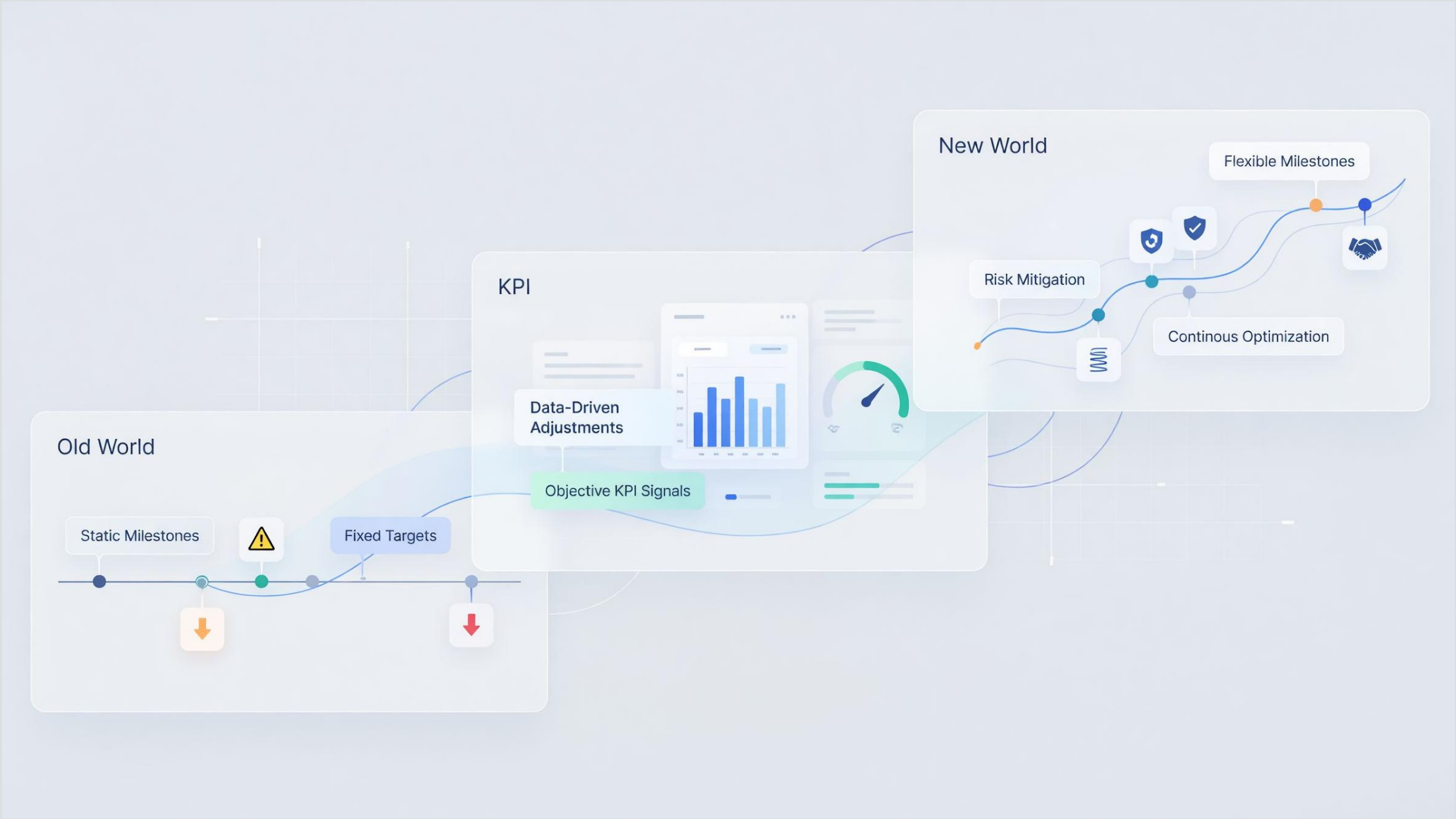
Milestones Timeline Planning



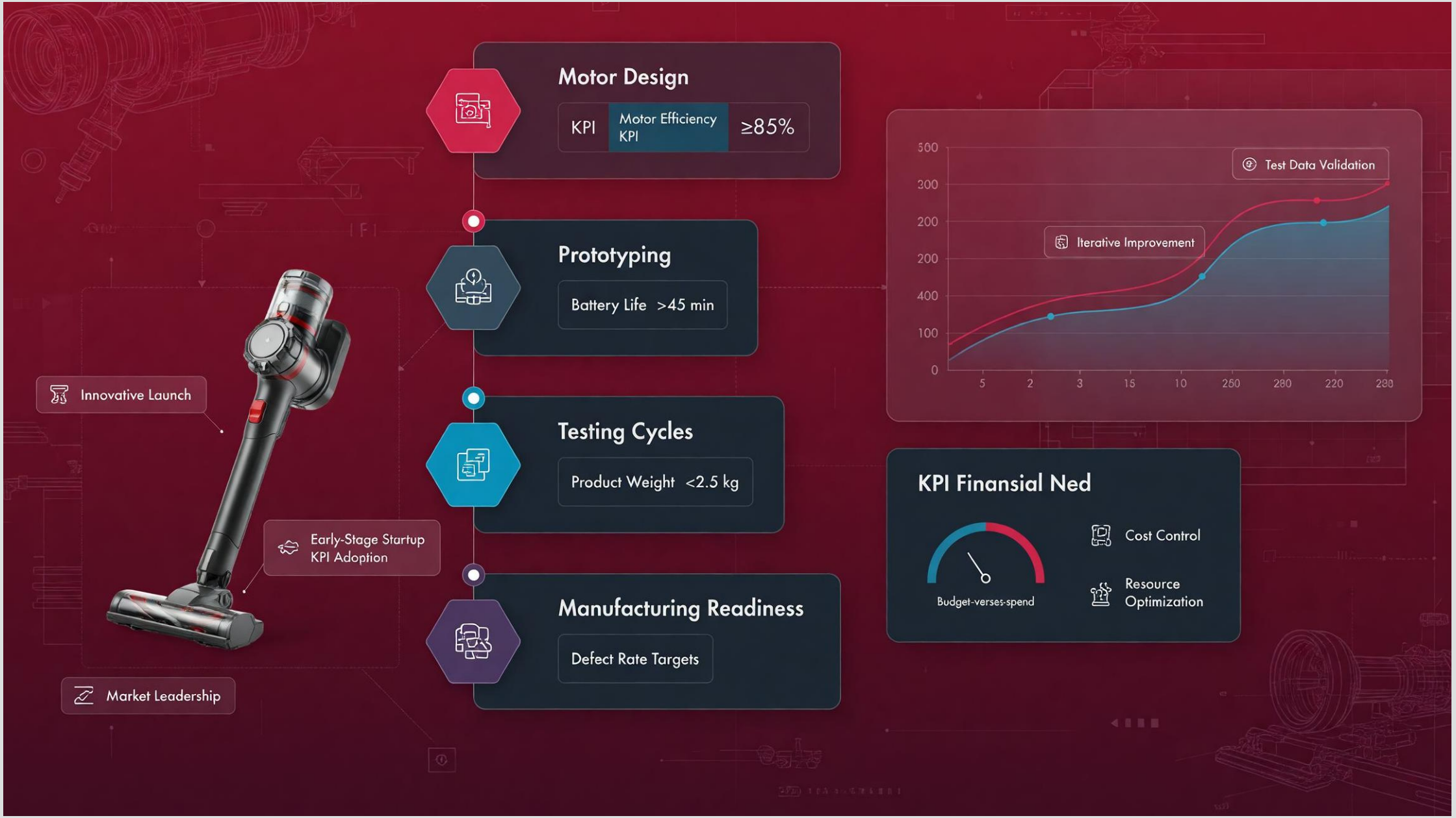
Monitoring Progress Using KPIs



Adjusting Milestones Based on Performance Data



Case Study: KPI and Milestone Setting in Technology Product Development



Chapter 5: Project Development Plan

Ir Prof Alan Lam, JP

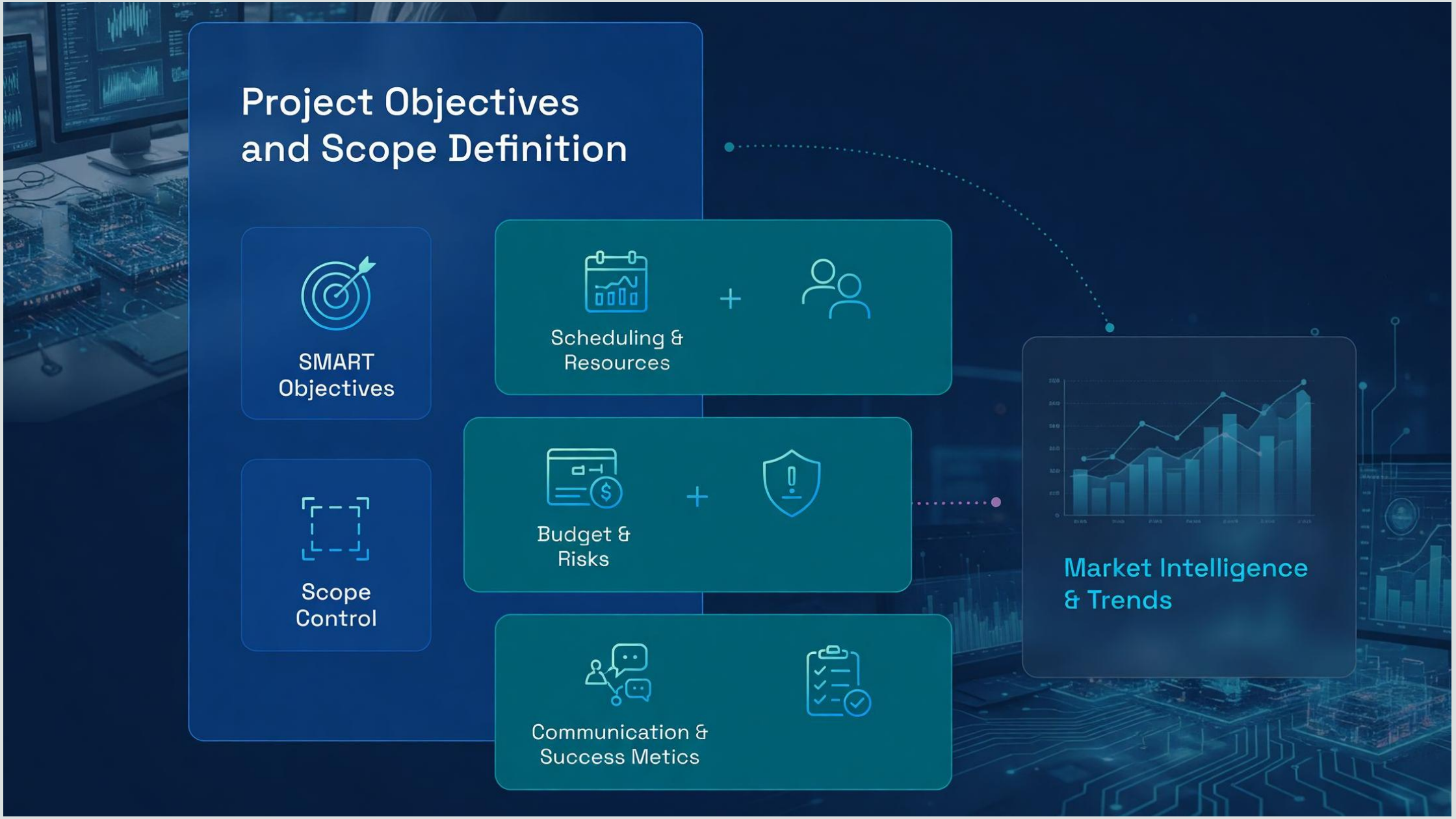


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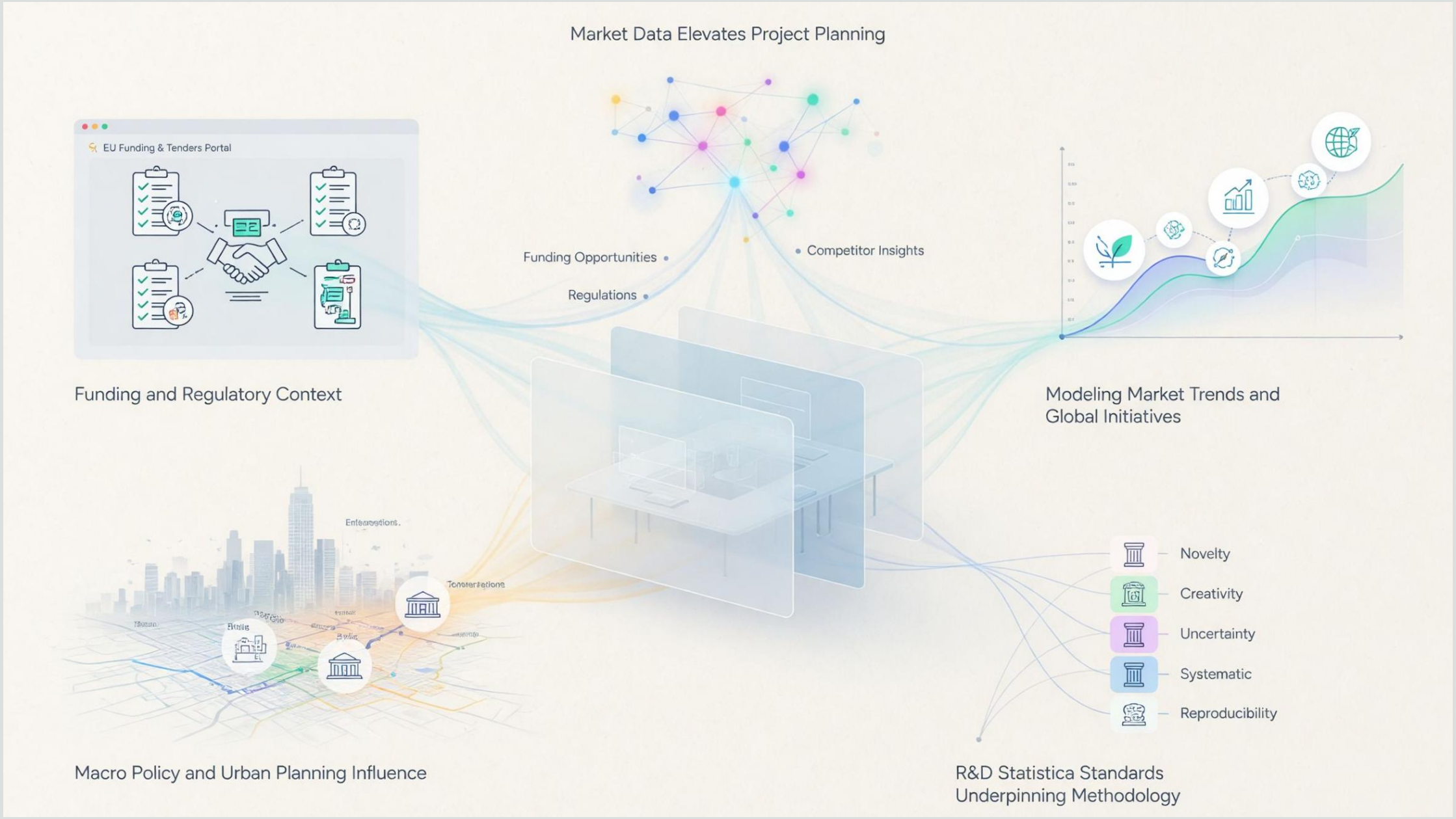
Introduction to Project Plan



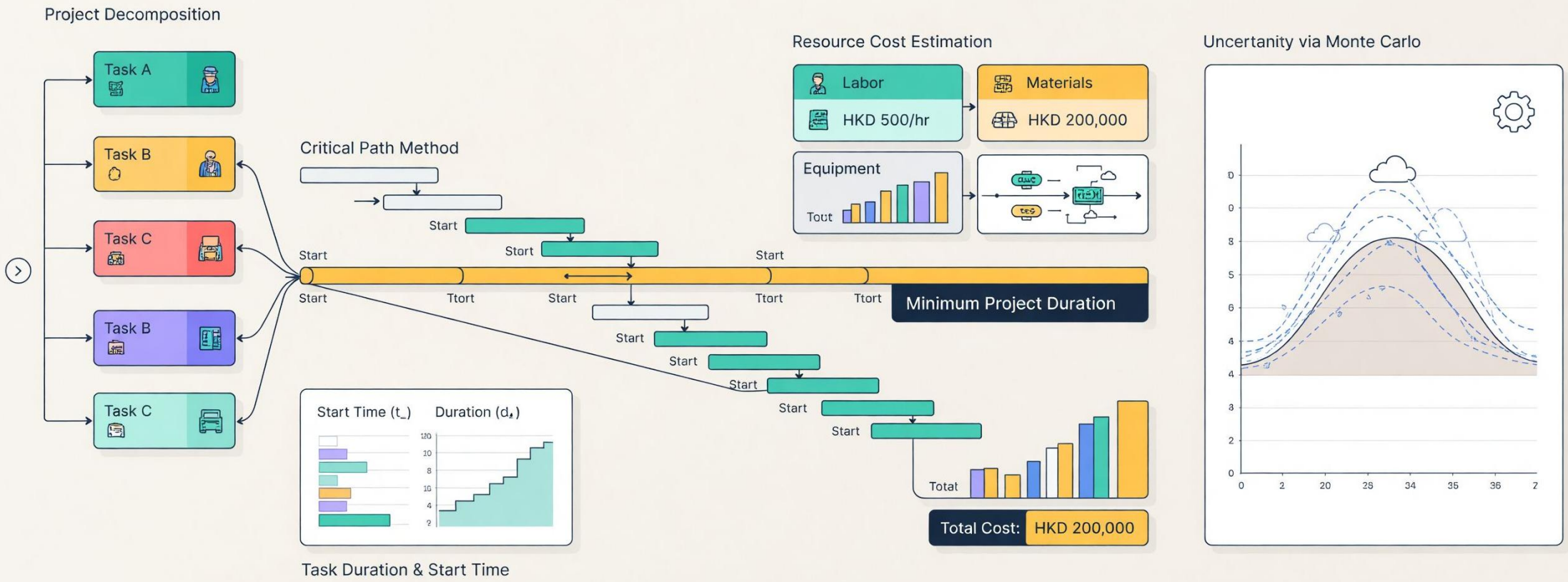
Components of a Project Development Plan



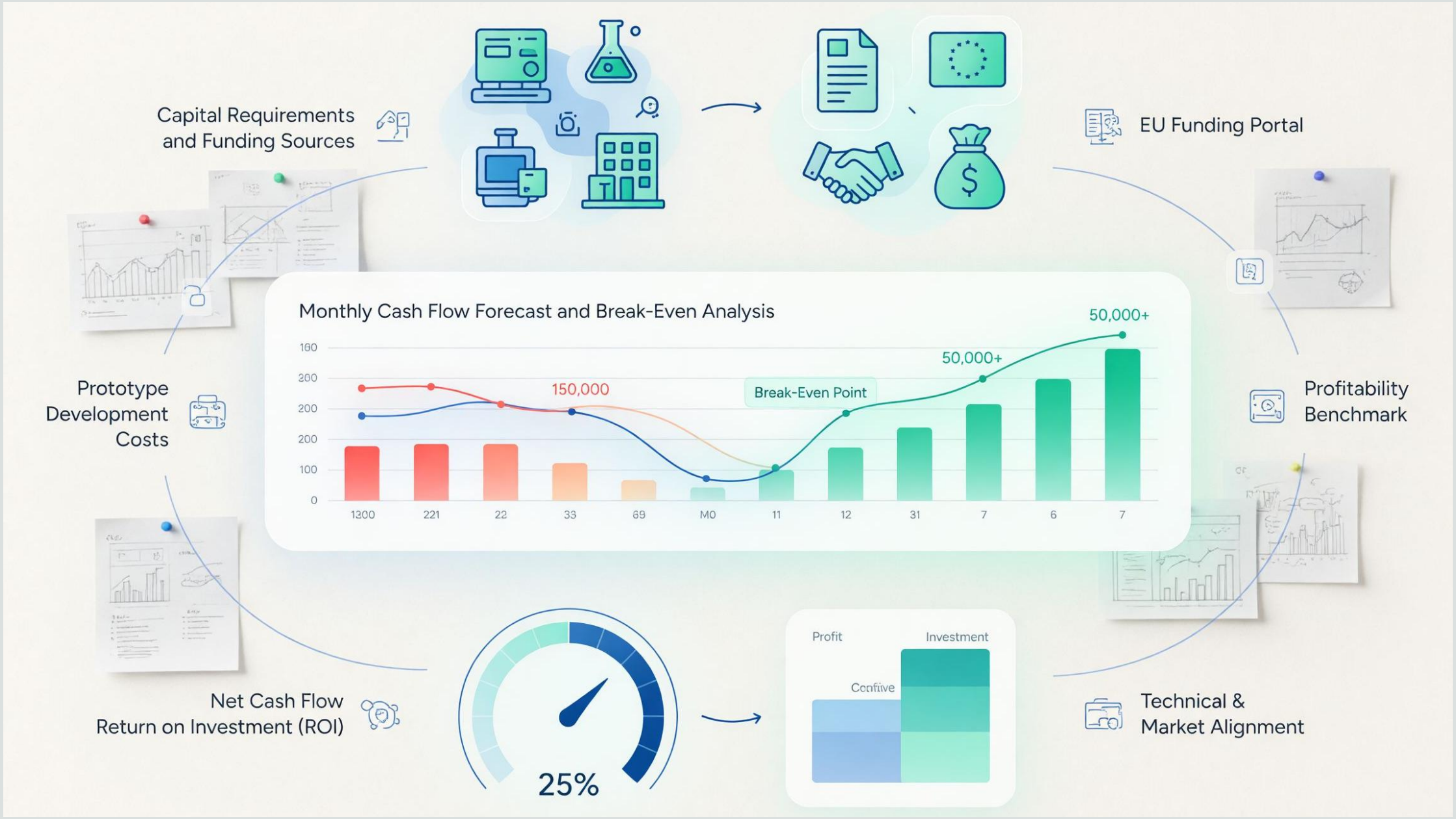
Market Data Integration into Project Planning



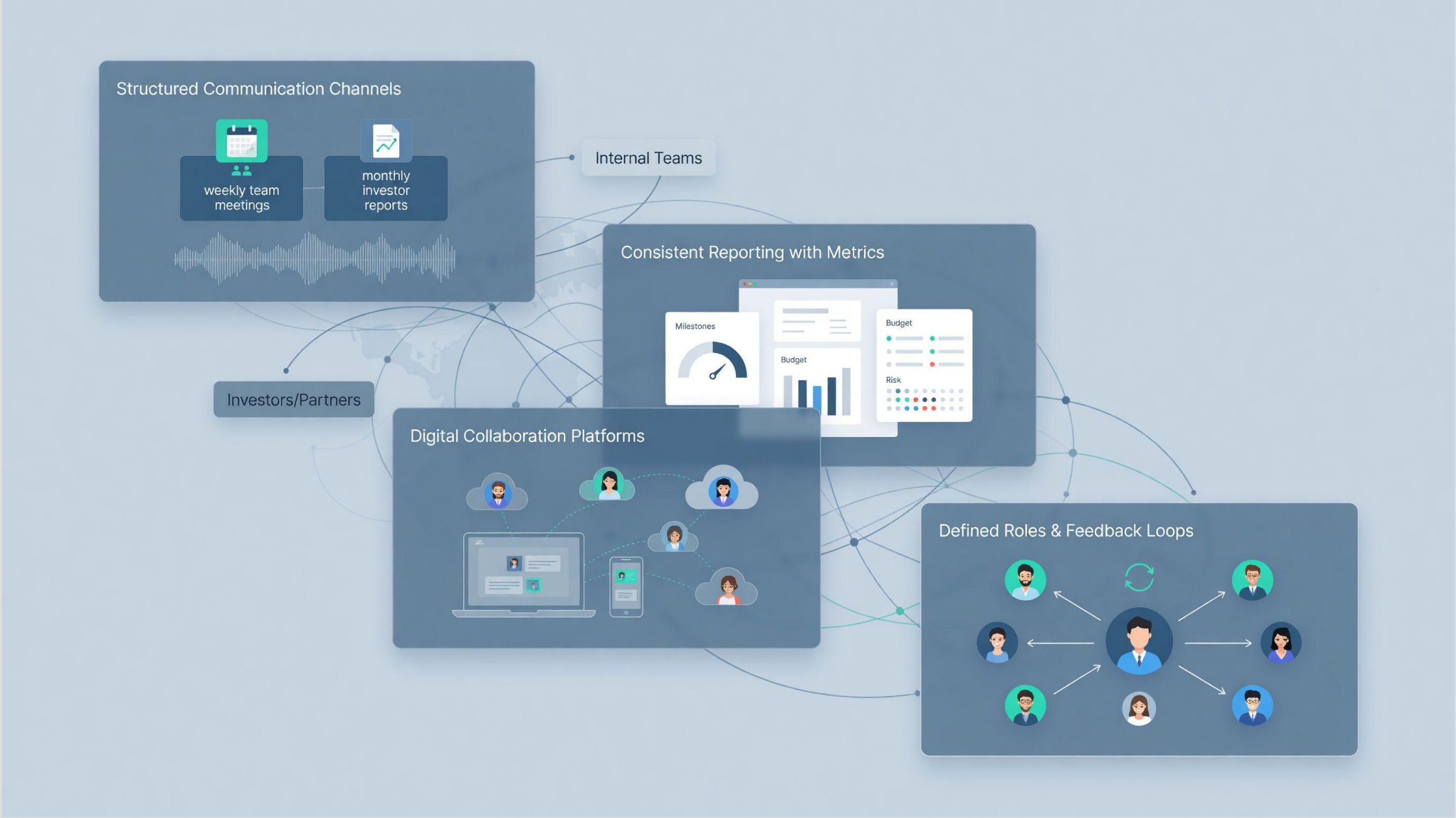
Resource and Time Estimation Techniques



Financial Planning in Project Plan



Communication and Reporting Structures



Project Plan Adaptation Based on Feedback



Framework of Project Development Plans

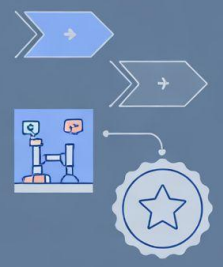
Market Data Analysis

Risk Management

Flexible Execution Strategies

Continuous Learning

- Iterative Design
- R&D Investment
- Market Pivot



- Compliance
- Technical & Financial
- Competitive Funding



- National Priorities
- Financial Feasibility
- Stakeholder Feedback



Chapter 6: Successful Example: Dyson for Successful Product Development

Ir Prof Alan Lam, JP

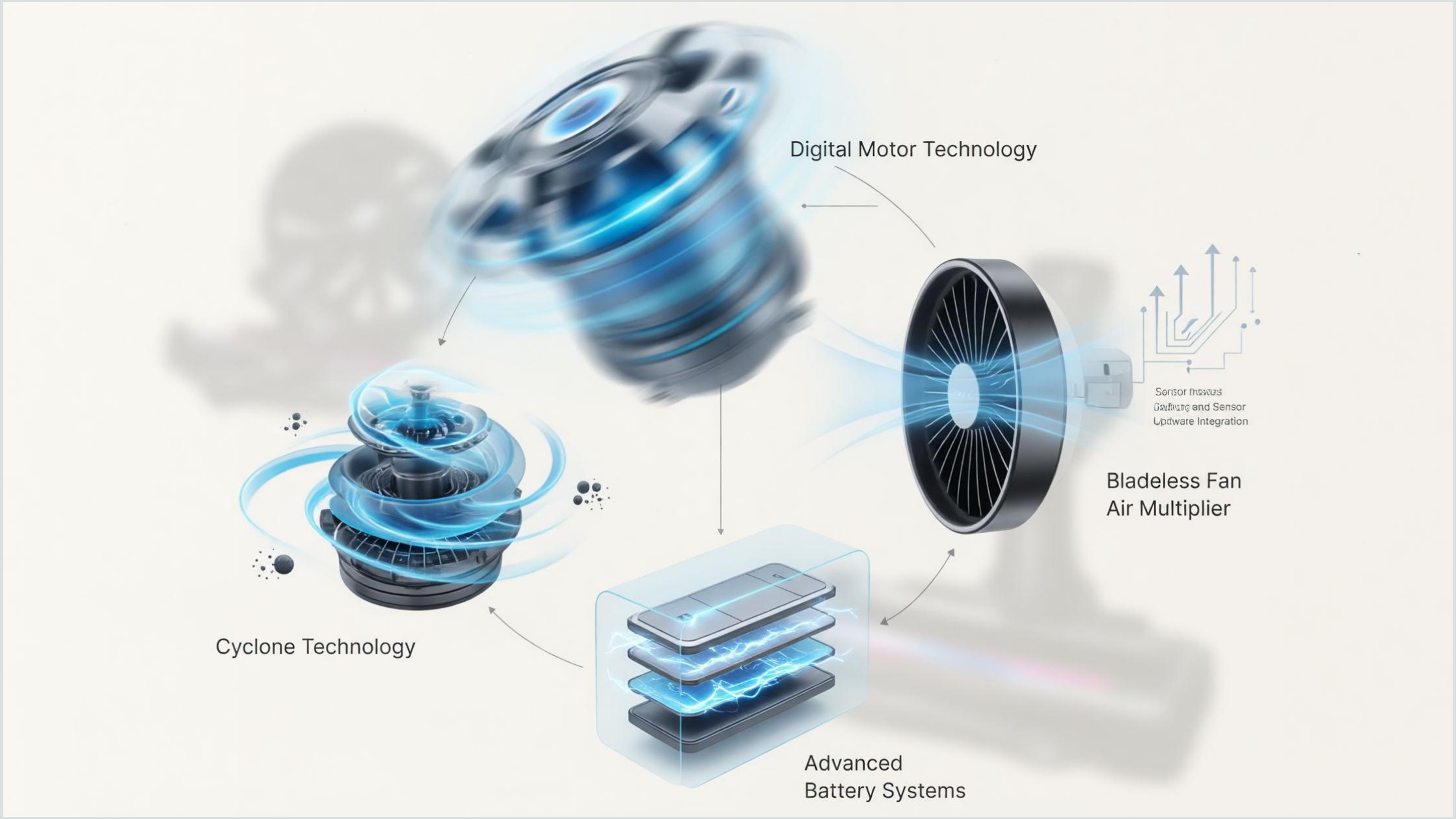


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Background and Foundation of Dyson's Product Development



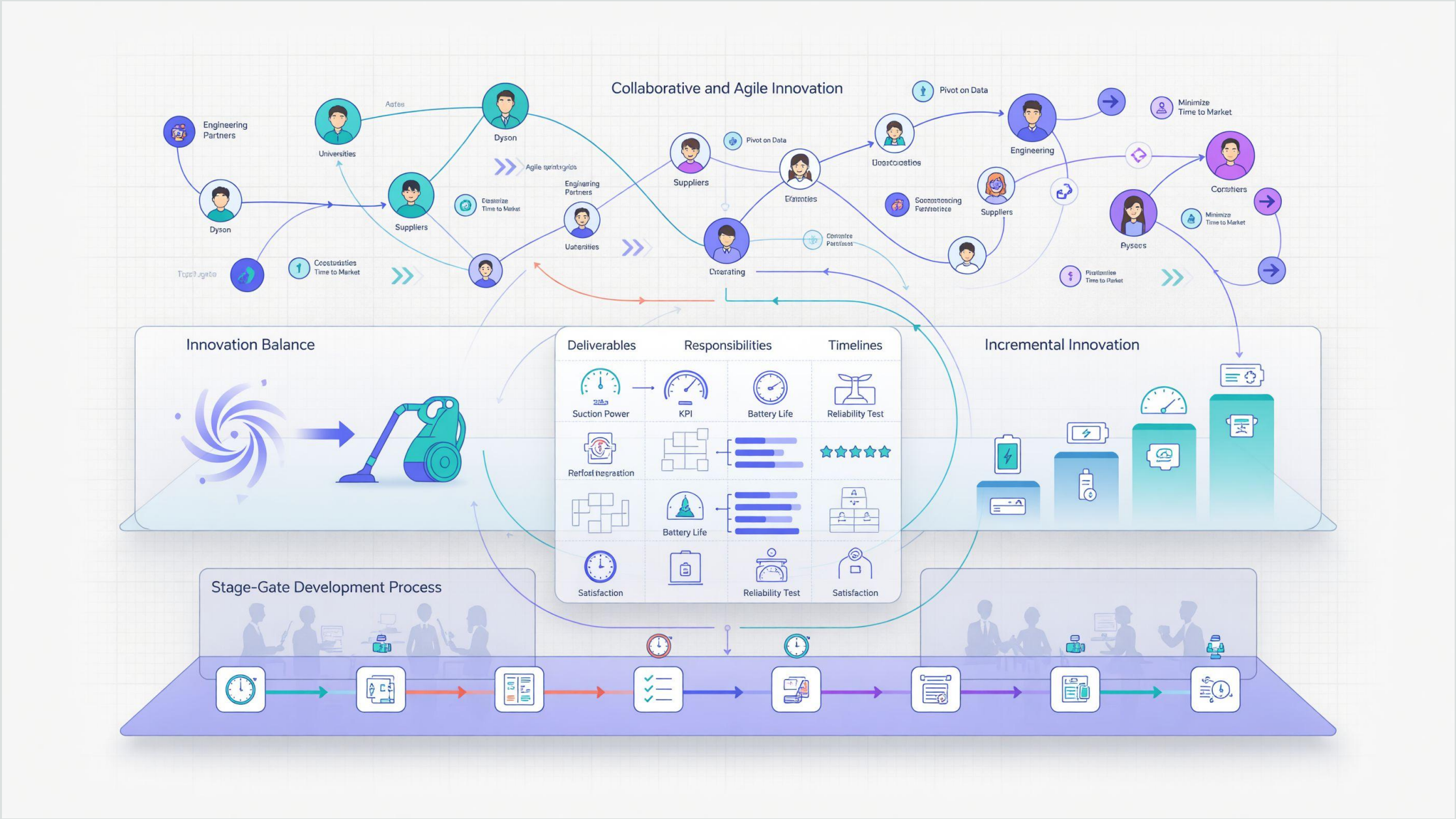
Innovative Technologies Implemented



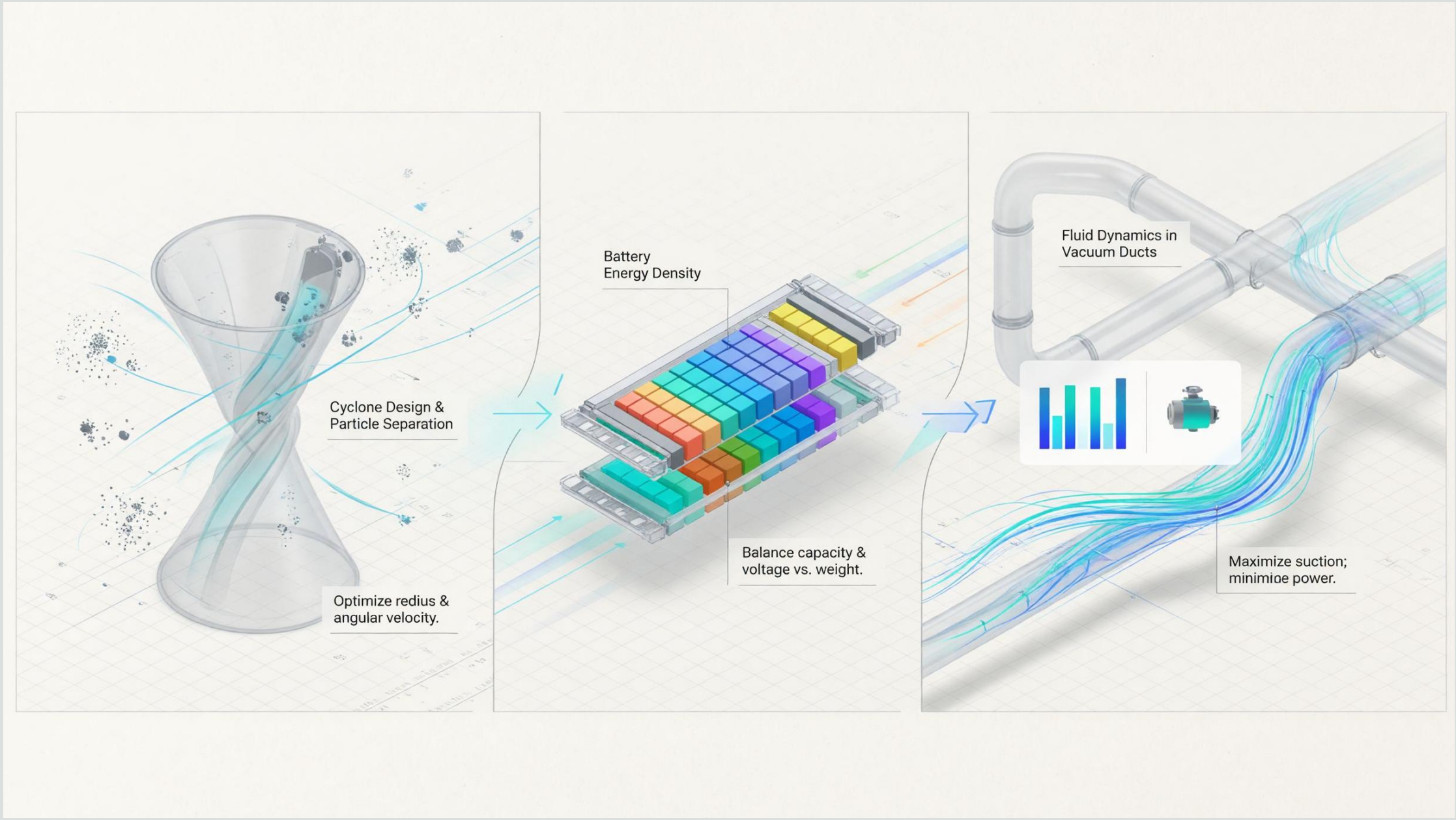
Market Analysis Driving Product Features



Product Development Strategies Employed



Technologies in Product Design Optimization



Commercialization Process and Market Reception Data



Lessons Learned and Best Practices for Technology-based Startups

Innovative User-Centric Solutions

User Pain Points & Breakthrough Tech

- Cyclonic Suction Innovation
- Category Redefined

Robust R&D and Engineering Practices

- Heavy R&D Investment
- Multiple Prototyping



- Optimized Design Risk

Hybrid Innovation Strategy



Collaborative and Adaptive Development

- Cross-Disciplonary Teams



Performance Tracking and Brand Building

- Tech Leadership
- KPI KPI
- Milestones & KPIs
- Consumer Trust

In-Class Discussion:

- Please based on the today's lecture to summarize what did you learn.
- Based on your project to work out WPC and your project plan with project scope and KPI for your MVP.
- Make one Slide about the Project milestone based on the WPC and the project development plan.



Reference:

- https://www.apm.org.uk/resources/what-is-project-management/
- https://www.investopedia.com/terms/p/project-management.asp
- https://www.projectmanager.com/guides/project-management
- https://www.usability.gov/what-and-why/project-management.html



Ir Prof Alan Lam, JP
Adjunct Professor

End of Presentation
Thank You